55(569.4)GOLD

THE GEOPHYSICAL INSTITUTE OF ISRAEL המכון הגיאופיסי לישראל
P.O.B. 2286, HOLON 58122, ISRAEL, TEL: 972-3-5576050, FAX: 972-3-5502925 רח' המשביר 1 חולון,

GEOLOGICAL SURVEY
12 MAY 1998

סקר אלקטרומגנטי לאיתור וניטור הפן-הבינימי באקויפר החוף

ינואר, 1998 דו"ח מס' 909/169/96

מאת

מרק גולדמן, דרור גלעד, מיכאל יזרסקי

הוכן עבור:

השרות ההידרולוגי - נציבות המים

1. הקדמה

המכון הגיאופיסי לישראל ביצע מדידות אלקטרומגנטיות בשיטת TDEM עבור השרות ההידרולוגי. במסגרת זו נערכו 60 מדידות באזור חוף הים התיכון מאלי סיני בדרום ועד קיסריה בצפון. חלק מהמדידות בוצע בתחנות נטור קבועות בשנים 1989, 1991, 1993 ו-1995 (ראה טבלה מס' 1, שבה מוצגת המדידה האחרונה שבוצעה באתר). בנוסף, הטבלה כוללת גם מידע כללי על המדידות כמו: מיקום, רום קרקע, גודל הלולאה ומרחק מהים.

במדידות חוזרות מטרת הסקר היתה לעקוב אחר התנודות בעומק הפן-הביני בזמן. התנודות באות לידי ביטוי בשינוי בעומק השכבה הגיאוחשמלית המוליכה ביותר בחתך (התנגדות חשמלית נמוכה מ-3 אוהם-מטר).

במדידות באתרים חדשים מטרת הסקר היתה בעיקר קביעת העומק עד שכבה (או שכבות) מוליכה/ות, לצורך מיפוי חדירת מי הים לאקויפר באזורים בהם אין קידוחי תצפית. בנוסף לכך, מספר מדידות בוצעו בקרבת קידוחי תצפית פן-ביני לצורכי כיול וביקורת מדידות ה-TDEM.

תחנות המדידה החדשות אוזנו ב-GPS דיפרנציאלי. הסטיה המירבית במיקום וברום הקרקע נמוכה מ-1 מ'. תחנות המדידה החוזרות אוזנו בעבר ע"י מודד עד כדי דיוק של 1 מ'. התחנות ליד קידוחי תצפית לא אוזנו, מקומן נקבע על פי נקודת ציון ורום הקידוח. יש לציין שרום הקידוח אוזן ע"י מודד אך מיקומו נקבע עפ"י מפה טופוגרפית, לכן נקודת הציון המקורבת נרשמה בטבלה ב-5 ספרות.

2. השיטה

שיטות העבודה בניטור ובאיתור הפן-הביני מתוארות בפרסומים ודו"חות מקצועיים קודמים (Goldman et al., 1988, רבינוביץ ואחרים, 1993). לפיכך נעבור ישירות לדיון על תוצאות הסקר.

3. תוצאות

תוצאות המדידות ופענוח נתוני השדה, בהם הובחנה יחידת התנגדות נמוכה מ-3 אוהם-מטר (המתאימה להתנגדות מי-ים), מוצגים בטבלה מס' 2 ובנספח לדו"ח זה. בנוסף מוצגות בטבלה תחנות TDEM בהן לא נמצאו מי ים ואלו שלא ניתנו לפענוח עקב רעש אלקטרומגנטי.

הנתונים המוצגים בטבלה כוללים רק חלק מתוצאות הפענוח (התוצאות המלאות מוצגות בנספח לדו"ח).

בטבלה הוגדרו שתי יחידות התנגדות נמוכה בחלק מהמדידות. היחידה העליונה מתאימה בדרך כלל למי הים באקויפר. היחידה התחתונה מתאימה בד"כ לחרסית הסקיה שבעומק. בחלק מהנקודות הובחנה יחידת התנגדות נמוכה אחת בלבד. כאשר היא נמצאה בחלק העליון של החתך הגיאולוגי שוייכה ליחידת התנגדות נמוכה עליונה, בעוד שכאשר היא נמצאה בחלק התחתון, שוייכה ליחידה התחתונה.

במספר נקודות מדידה נמצאה יחידת התנגדות גבוהה יחסית, המונחת על-גבי יחידת התנגדות נמוכה ממנה. תופעה זו מצביעה על תחום ערבוב של מים מליחים הנמצאים מעל מי-ים. במקרים אלה ערכי התנגדויות ועומקים של שתי השכבות הללו מוצגים בטבלה 2 באותה עמודה, כאשר הם מחולקים על-ידי סימן "/".

רשימת פרסומים

רבינוביץ, ב., גולדמן, מ., רבינוביץ, מ., רונן, ע., 1993. סקר אלקטרומגנטי לאיתור הפן הביני באקויפר החוף, דו"ח מס' 735/296/92, המכון הגיאופיסי.

Goldman, M., Arad, A., Kafri, U., Gilad, D. and Melloul, A., 1988. Detection of freshwater/ seawater interface by the Time Domain Electromagnetic (TDEM) method in Israel, 10th SWIM, Ghent.

Table 1

TDEM station HS...	Location	Loop size (m)	Easting	Northing	Elevation (m)	Distance from the sea (m)	Previous station HS...	Comments
403	Ashdod	100x100	118446	140421	4.0	330		GPS
404	Ashdod (N)	100x100	118275	140503	2.5	150		GPS
405	Ashdod	100x100	117014	135989	16.0	900		GPS
406	Ashdod	50x50	115916	135178	4.5	230		GPS
407	Ashdod	100x100	115257	133498	13.5	350		GPS
408	Ashdod	50x50	115079	133542	0.5	190		GPS
409	Ashdod	100x100	114314	13166	12.0	340	269 (*)	
410	Ashqelon	100x100	108380	119340		820		noisy (NI)
411A	Rishpon	100x100	13320	17925	41.0	830	241 (*)	well 38/B
412	Rishpon	50x100	133440	179360		1000	245 (*)	noisy (NI)
413	Rishpon (beach)	100x100	132645	179682	41.5	170	243 (*)	
414	Rishpon	100x100	133492	179517	37.5	1000		GPS, noisy
415	Ga'ash	150x150	134185	18286	25.5	750	244 (*)	
416	Ashqelon	100x100	107224	11758	41.5	1250		
417	Ashqelon	100x100	107264	118085	38.0	1000	273 (*)	
418	Ashqelon	100x100	107212	117838	40.5	1130		GPS
419	Bat-Yam	100x100	125427	15719	7.5	370		GPS
420	Bat-Yam	45x85	126746	158051	31.5	1400		GPS, noisy
421	Bat-Yam	50x50	12730	15970		1500		noisy (NI)
422	Palmakhim	100x100	12161	14812	13.0	180	332	well 22/B
423	Sde-Dov	50x50	12943	16912	4.0	70	235	well 33/O
424	Hertzliya (S)	100x100	130991	17328	19.0	290		GPS, noisy
425	Kiryat Khinukh	100x100	13025	17117	19.0	180	236 (*)	well 340/L
426	Hertzliya (S)	100x100	131177	17323	22.5	480		GPS, noisy
427	Hertzliya	100x100	13158	17462	14.0	490	239 (*)	well 36/A
428	Hertzliya	50x50	13125	17468	9.0	180	238 (*)	well 36/O, noisy
429	Nof-Yam	75x75	132234	17748	20.5	500	242 (*)	noisy
430	Yakum	100x100	134805	184085	18.5	1000		GPS
431	Hof Zikim	100x100	103116	113504	2.5	180		GPS
432	Hof Zikim	100x100	103363	113428	3.0	450		GPS
433	Zikim	100x100	103954	114487	5.0	350	276 (*)	
434	Zikim	100x100	104122	114378	11.0	550	275 (*)	
435	Netanya (S)	100x100	135505	188555	41.5	600	259 (*)	
436	Netanya (S)	100x100	13628	190674	47.0	820		GPS
437	Netanya (S)	100x100	135898	190983	40.0	430		GPS
438	Rishpon	100x100	133824	179514	45.0	1330		GPS
439	Netanya (N)	80x80	137297	194806	20.5	900		GPS, noisy
440	Beitan Aharon	100x100	137468	19685	13.0	700		GPS
441	Bet Yanai	50x50	137888	199255	6.0	520	248 (*)	noisy
442	Givat Olga	100x100	139081	204636	15.0	450	257 (*)	GPS
443	Givat Olga	100x100	139304	204138	14.0	900		GPS
444	Takhanat Koakh Hadera	100x100	13980	20770	6.5	550		well 53/B

Table 1 (cont'd)

TDEM station HS...	Location	Loop size (m)	Easting	Northing	Elevation (m)	Distance from the sea (m)	Previous station HS...	Comments
445	Takhanat Koakh Hadera	100x100	13990	207113	12.0	800		GPS
446	Alei Sinai	100x100	10220	11120		550		noisy (NI)
447	Poleg, Hof Hayarok	100x100	13467	187122	8.5	160	247(*)	
448	Poleg	100x100	13482	18710		310		well 42/A
449	Poleg	100x100	135057	18707	9.5	540		GPS
450	Ga'ash	100x100	134536	182581	29.5	1100		GPS, noisy
451	Netanya (S)	100x100	135638	188408	34.5	750		GPS
452	Betan Aharon	70x70	137651	196937	14.5	890		GPS
453	Hofit	50x50	13803	19880	9.0	800		well 48/BB
454	Mikhmoret	100x100	13841	20023	6.0	900		well 49/A
455	Katza (Ashqelon)	100x100	105293	115825	14.0	650		GPS
456	Katza (Ashqelon)	100x100	105579	115881	21.0	880		GPS
457	Ashqelon (S)	100x100	104507	114266	22.5	940		GPS
458	Ashdod	100x100	115492	133317	14.5	630		GPS, noisy
459	Caesaria	200x200	140302	211513	7.0	400	71	GPS
460	Mikhmoret	100x100	139275	202471	21.5	1150		GPS
461	Mikhmoret	100x100	138567	202743	14.0	440	256(*)	GPS
462	Mikhmoret	100x100	138108	200442	4.0	550	254(*)	GPS

NI = not interpreted

(*) = permanent TDEM monitoring station (barrel)

Table 2 Resistivities and depths to the low resistivity unit(s) associated with seawater intrusion and/or the Saqiya clays

TDEM station HS...	Hydrological strip	Upper layer (s)		Lower layer (s)		Comments
		Resistivity (ohm-m)	Depth (m)	Resistivity (ohm-m)	Depth (m)	
403	118	3.0	56	1.8	174	High resistivity
404	118	1.6	22	-	-	
405	116	-	-	-	-	No interface
406	115	1.2/0.9	27/32	-	-	
407	114	2.5	46	2.9	155	Two subaquifers
408	114	1.1	27	1.2	145	Two subaquifers
409	113	3.1/1.8	50/71	-	-	
410	106	-	-	-	-	Noisy, not interpeted
411A	138	1.6	91	1.5	229	
412	138	-	-	-	-	Noisy, not interpeted
413	138	1.4	73	-	-	
414	138	1.1	95	1.3	199	Noisy
415	140	1.2	81	2.1	159	
416	105	-	-	2.2	204	
417	105	1.3	113	1.2	209	
418	105	2.1	123	1.4	204	
419	127	2.3	50	2.6	215	
420	128	3.8/0.4	33/72	-	-	Noisy, not reliable
421	128	-	-	-	-	Noisy, not interpeted
422	122	2.8/1.6	41/63	1.0	216	
423	133	1.3	17	1.3	53	Two subaquifers
424	135	-	-	1.0	195	
425	134	2.7	46	-	-	Noisy
426	135	-	-	1.8	192	Noisy
427	136	2.9	63	-	-	
428	136	1.1	33	-	-	Noisy
429	137	1.7	61	-	-	Noisy
430	141	-	-	2.4	157	Two subaquifers
431	102	3.0	32	1.6	148	High resistivity
432	102	-	-	1.3	167	Lower subaquifer
433	103	1.5	44	1.5	143	Two subaquifers
434	103	3.3/2.0	49/77	1.2	164	Two subaquifers
435	143	2.0	76	1.6	209	
436	144	-	-	1.3	205	
437	144	1.4	101	1.7	226	
438	138	-	-	2.8	182	
439	146	-	-	-	-	No interface, noisy
440	147	1.7	58	2.4	164	
441	148	1.2	53	-	-	Noisy
442	151	1.4	50	1.3	148	
443	151	1.5	50	1.2	153	
444	153	2.1	46	-	-	

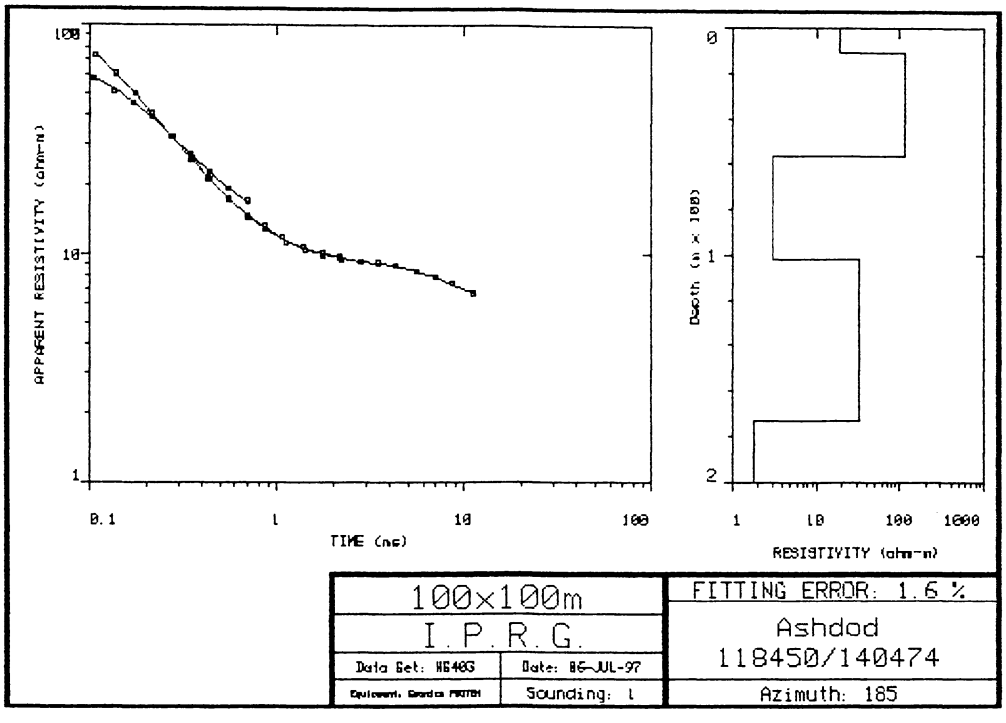
Table 2 (cont'd)

TDEM station HS...	Hydrological strip	Upper layer (s)		Lower layer (s)		Comments
		Resistivity (ohm-m)	Depth (m)	Resistivity (ohm-m)	Depth (m)	
445	153	-	-	2.3	103	
446	101	-	-	-	-	Noisy, not interpreted
447	142	1.5	33	-	-	
448	142	1.4	42	1.3	181	
449	142	1.3	53	1.4	175	
450	140	-	-	2.2	143	Noisy
451	143	-	-	2.0	167	
452	147	1.8	62	2.3	171	
453	148	1.3	56	-	-	
454	149	1.5	52	2.0	140	
455	104	1.8	66	-	-	
456	104	1.7	89	1.1	190	
457	103	2.2	80	1.1	200	
458	114	-	-	2.1	138	Lower subaquifer, noisy
459	155	1.5	53	2.2	108	
460	150	2.5	65	1.4	169	
461	150	1.4	48	-	-	
462	149	1.4	44	-	-	

נספח

HS403

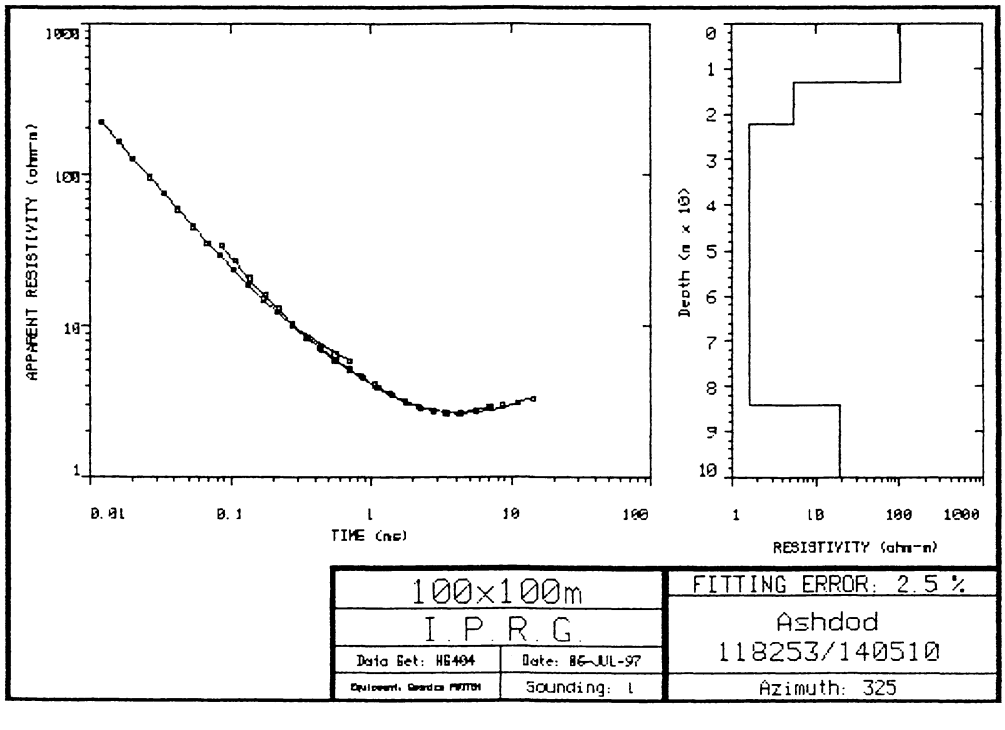
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.91961E+01	1.08589E+01
2	1.12636E+02	4.52559E+01
3	3.02895E+00	4.57761E+01
4	3.17397E+01	7.16431E+01
5	1.78282E+00	



100x100m	FITTING ERROR: 1.6 %
I.P.R.G.	Ashdod 118450/140474
Data Set: HS403 / Date: 06-JUL-97	
Equipment: Geotron PROTEM / Sounding: 1	Azimuth: 185

HS404

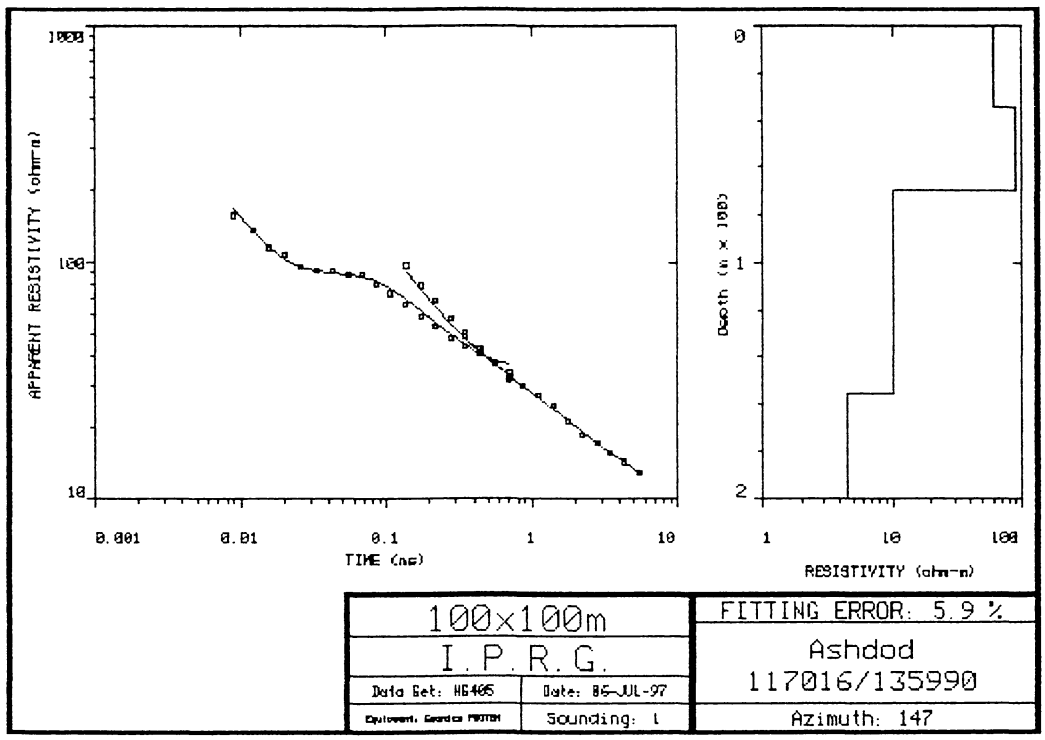
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.04545E+02	1.31911E+01
2	5.31156E+00	9.04903E+00
3	1.63247E+00	6.20176E+01
4	1.90813E+01	



100x100m	FITTING ERROR: 2.5 %
I.P.R.G.	Ashdod 118253/140510
Data Set: HS404 / Date: 06-JUL-97	
Equipment: Geotron PROTEM / Sounding: 1	Azimuth: 325

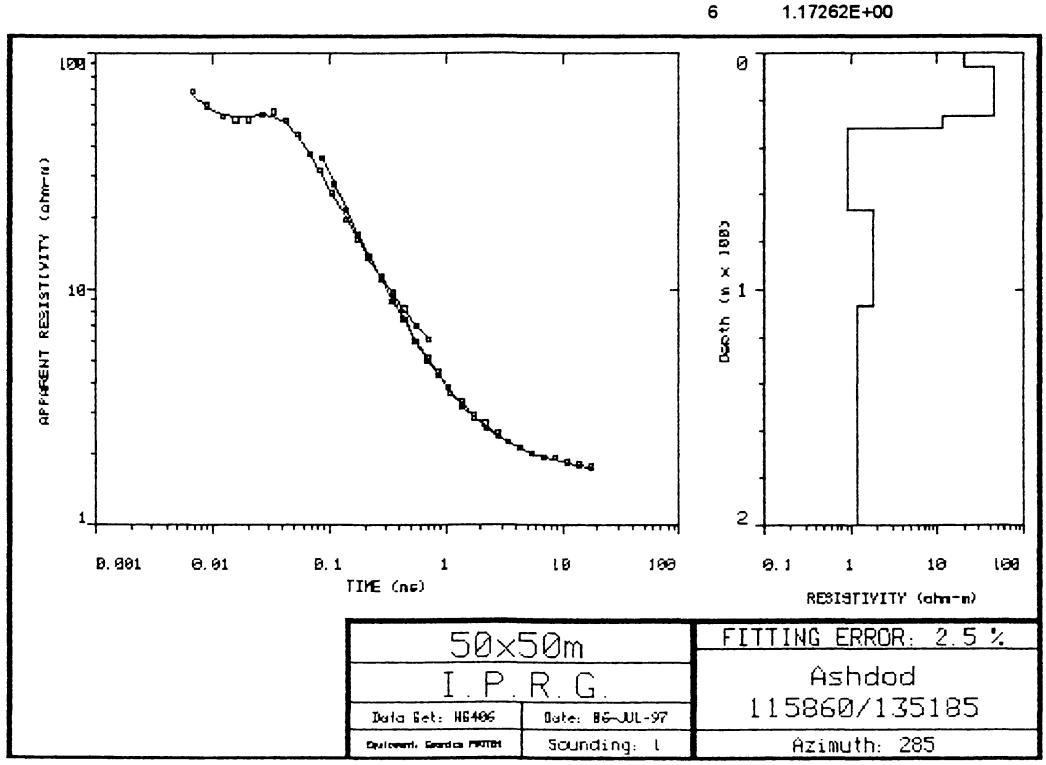
HS405

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	6.10805E+01	3.44193E+01
2	8.91458E+01	3.52292E+01
3	1.02593E+01	8.61949E+01
4	4.61726E+00	



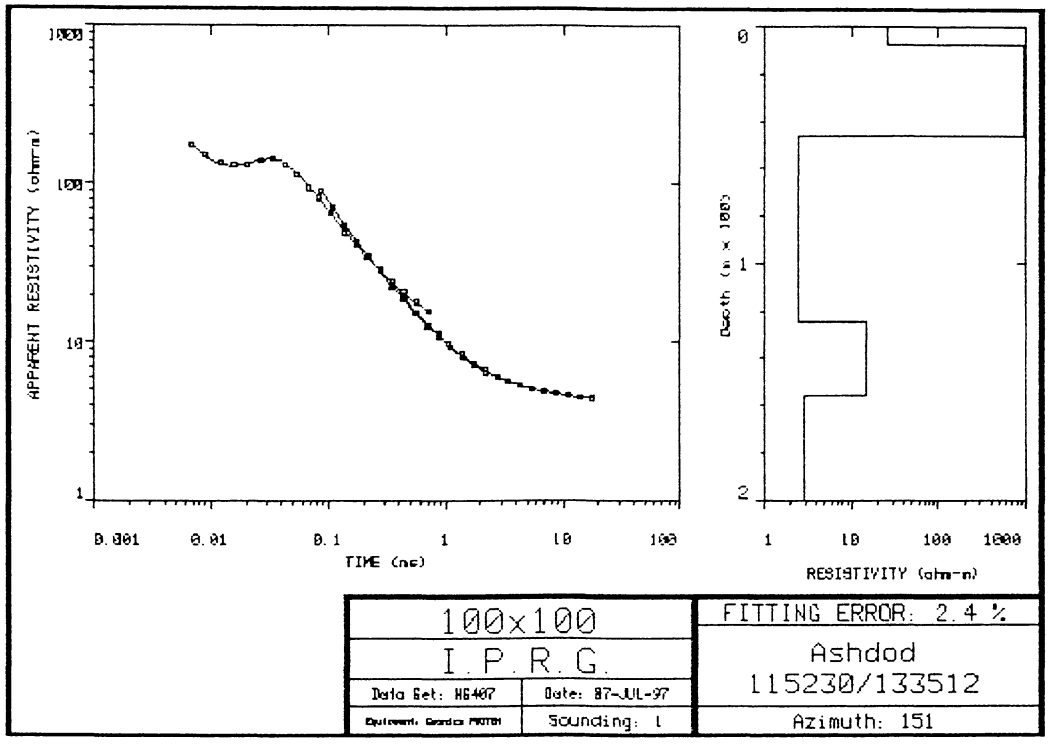
HS406

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.01745E+01	5.56811E+00
2	4.53367E+01	2.12742E+01
3	1.17822E+01	5.25125E+00
4	9.14608E-01	3.43907E+01
5	1.81204E+00	4.06367E+01
6	1.17262E+00	



HS407

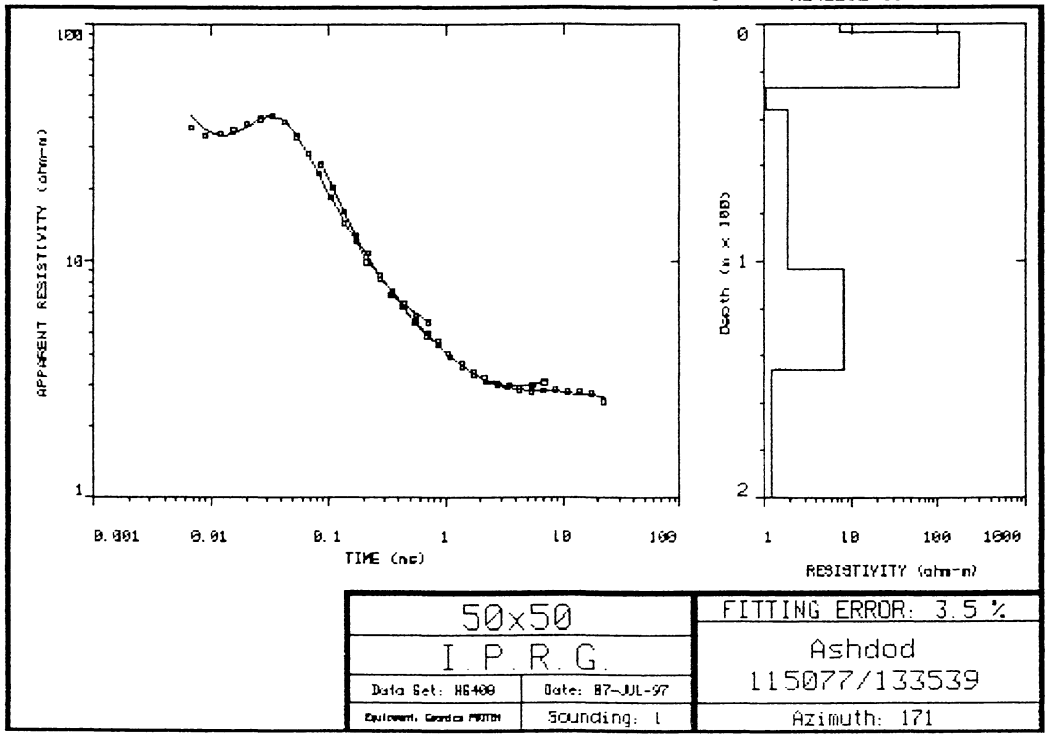
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.61385E+01	7.54288E+00
2	9.66251E+02	3.91088E+01
3	2.51584E+00	7.75688E+01
4	1.44678E+01	3.16444E+01
5	2.90254E+00	



100x100	FITTING ERROR: 2.4 %
I.P.R.G.	Ashdod 115230/133512
Data Set: HS407 / Date: 07-JUL-97	
Equipment: Geodica PROTEM / Sounding: 1	Azimuth: 151

HS408

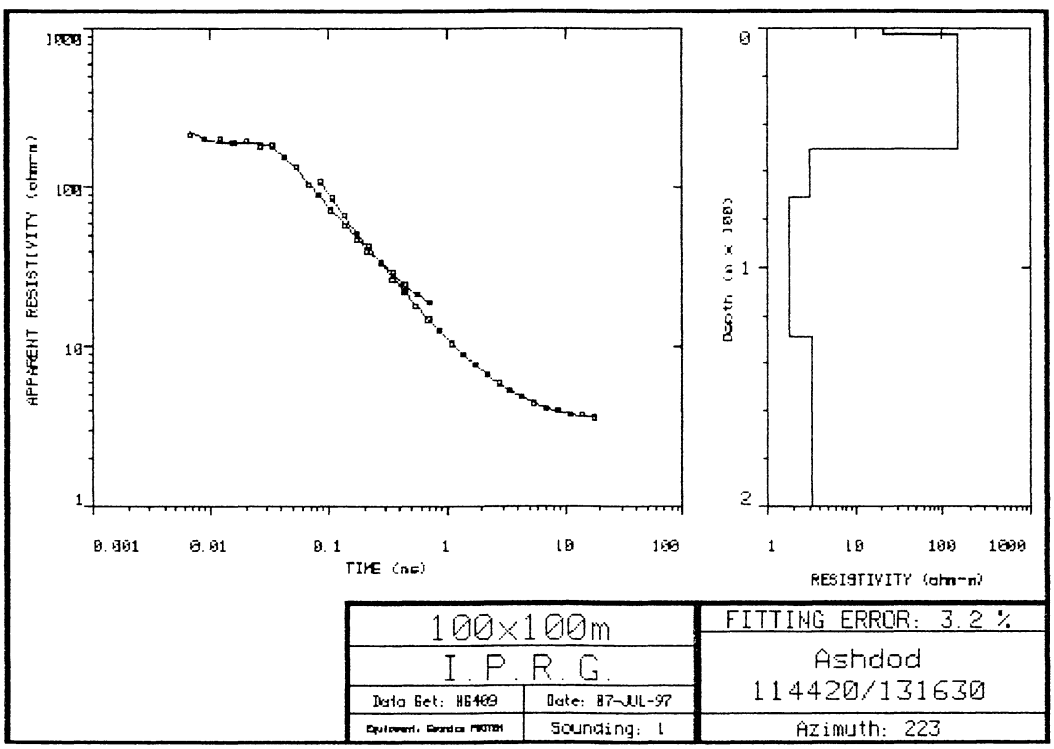
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	7.28386E+00	3.38199E+00
2	1.76816E+02	2.39394E+01
3	1.08489E+00	8.80064E+00
4	1.93747E+00	6.69621E+01
5	8.05217E+00	4.24782E+01
6	1.24225E+00	



50x50	FITTING ERROR: 3.5 %
I.P.R.G.	Ashdod 115077/133539
Data Set: HS408 / Date: 07-JUL-97	
Equipment: Geodica PROTEM / Sounding: 1	Azimuth: 171

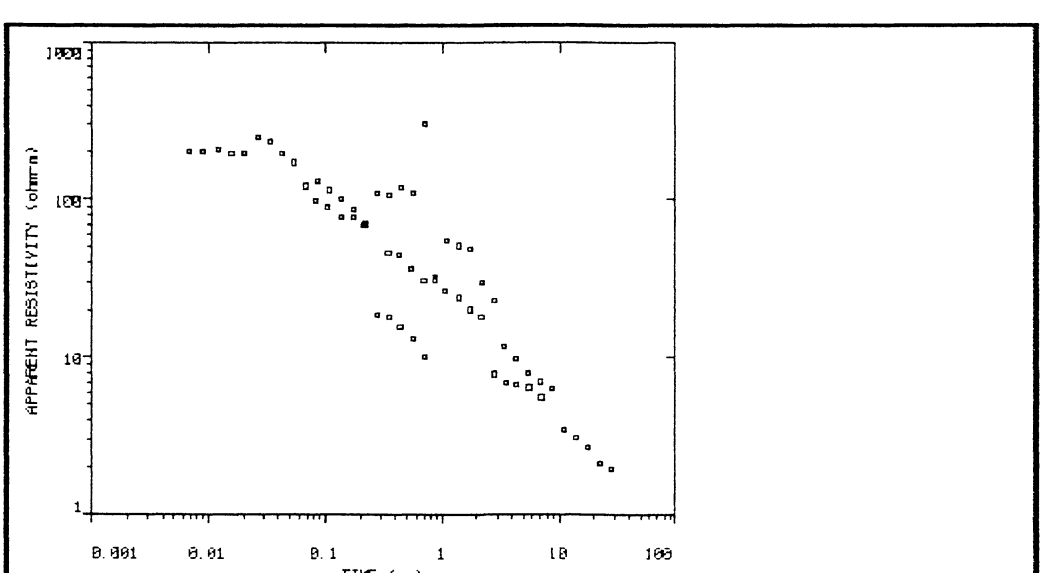
HS409

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.14358E+01	2.26655E+00
2	1.53331E+02	4.80515E+01
3	3.06440E+00	2.07448E+01
4	1.81673E+00	5.77159E+01
5	3.19146E+00	



100×100m	FITTING ERROR: 3.2 %
I.P.R.G.	Ashdod
Data Set: HS409 / Date: 07-JUL-97	114420/131630
Equipment: Geotics PROTEM / Sounding: 1	Azimuth: 223

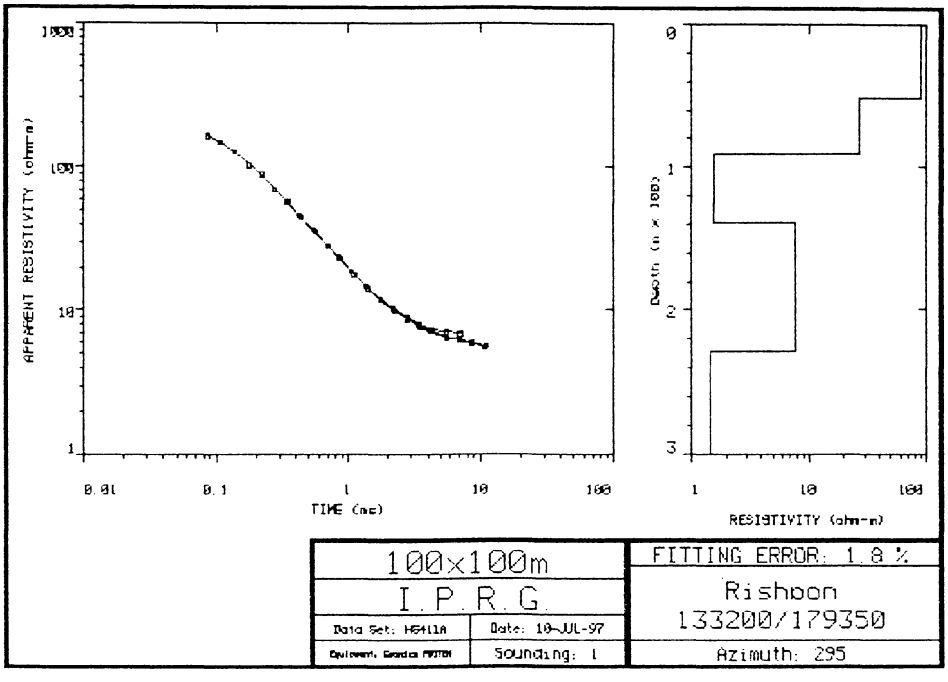
HS410



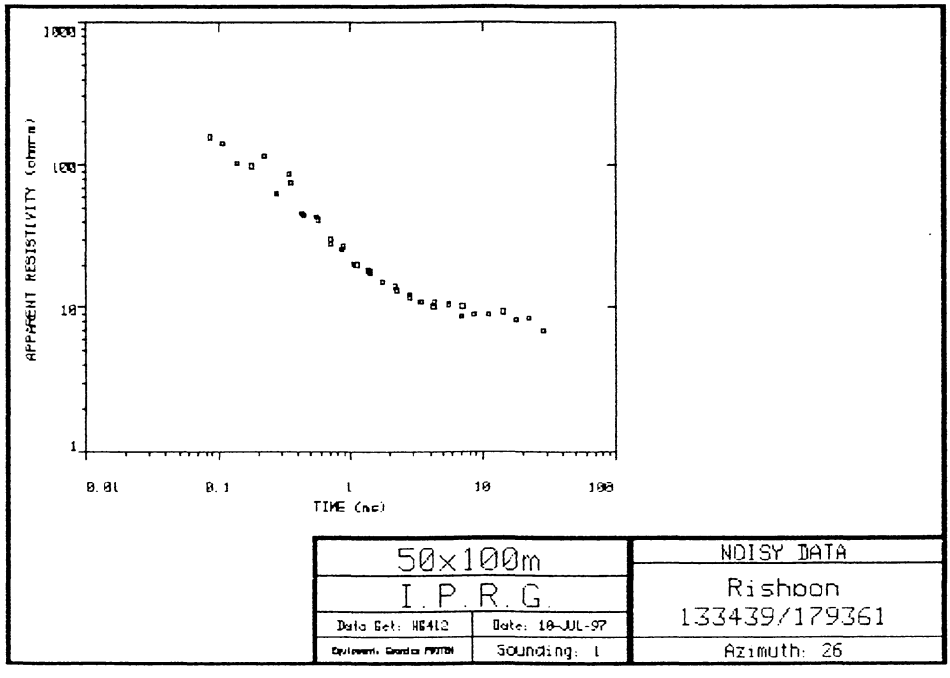
100×100m	NOISY DATA
I.P.R.G.	Ashqelon
Data Set: HS410 / Date: 07-JUL-97	108380/119340
Equipment: Geotics PROTEM / Sounding: 1	Azimuth: 360

HS411a

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	9.00079E+01	5.22092E+01
2	2.72401E+01	3.89414E+01
3	1.57070E+00	4.77025E+01
4	7.67975E+00	9.00993E+01
5	1.48528E+00	

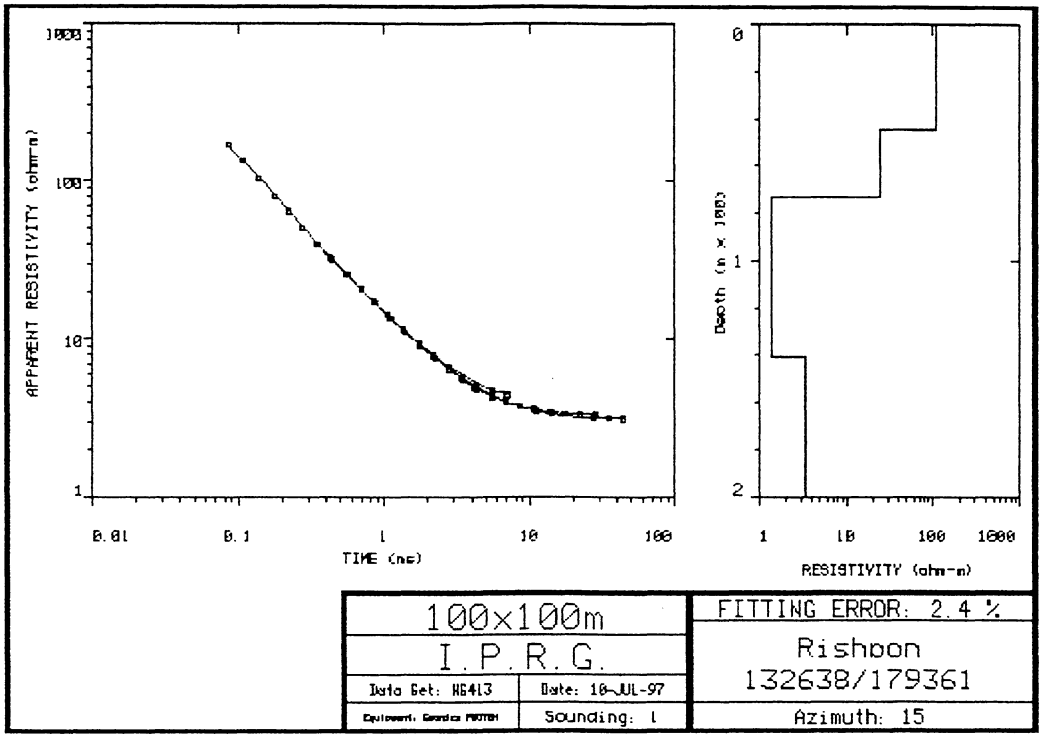


HS412



HS413

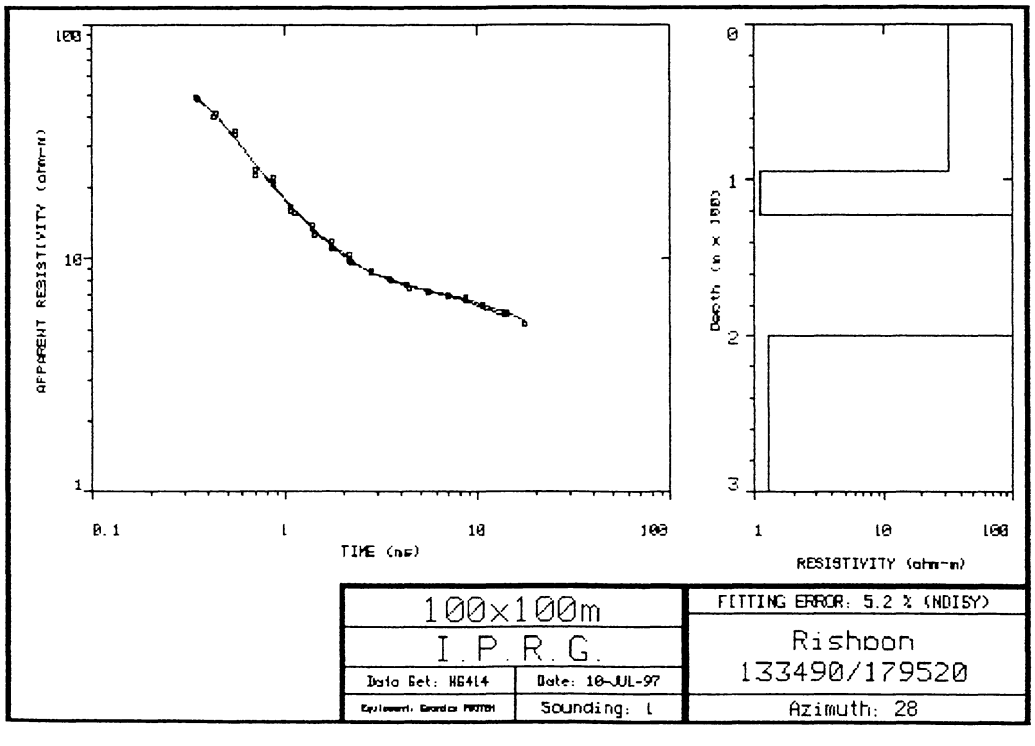
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.08565E+02	4.41891E+01
2	2.47204E+01	2.90689E+01
3	1.37505E+00	6.68566E+01
4	3.40815E+00	



100x100m	FITTING ERROR: 2.4 %
I.P.R.G.	Rishoon 132638/179361
Data Set: HS413 / Date: 10-JUL-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 15

HS414

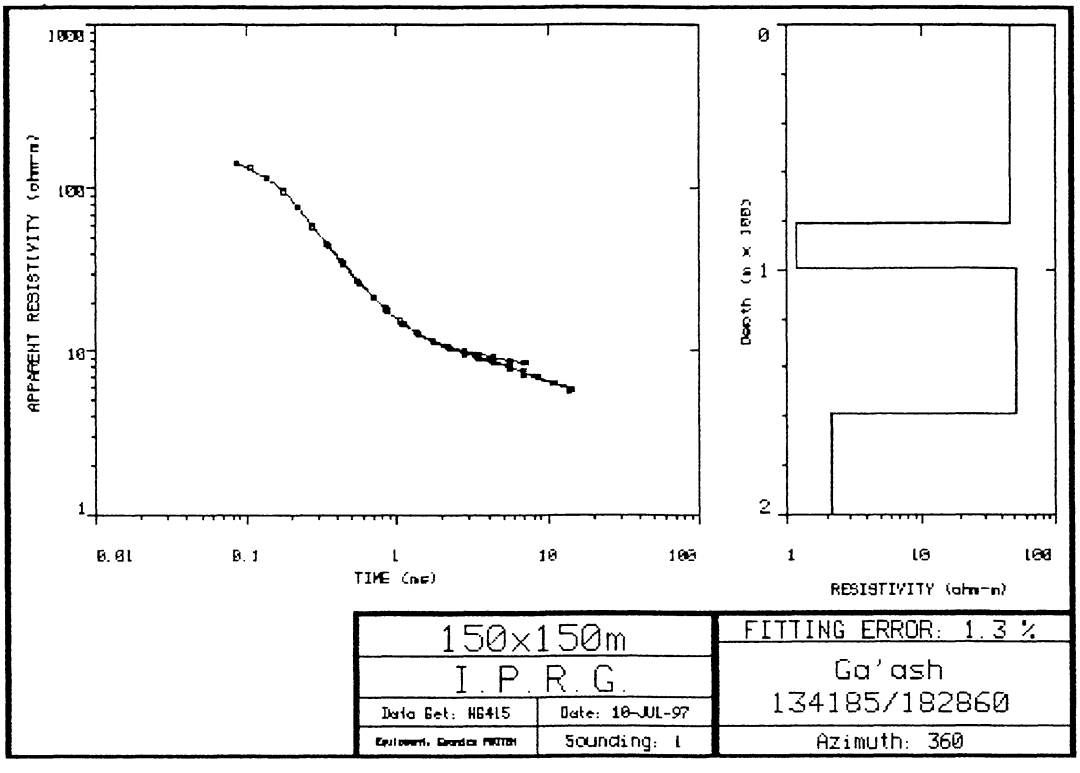
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	3.21224E+01	9.50752E+01
2	1.11716E+00	2.68127E+01
3	9.90782E+01	7.70811E+01
4	1.26446E+00	



100x100m	FITTING ERROR: 5.2 % (NOISY)
I.P.R.G.	Rishoon 133490/179520
Data Set: HS414 / Date: 10-JUL-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 28

HS415

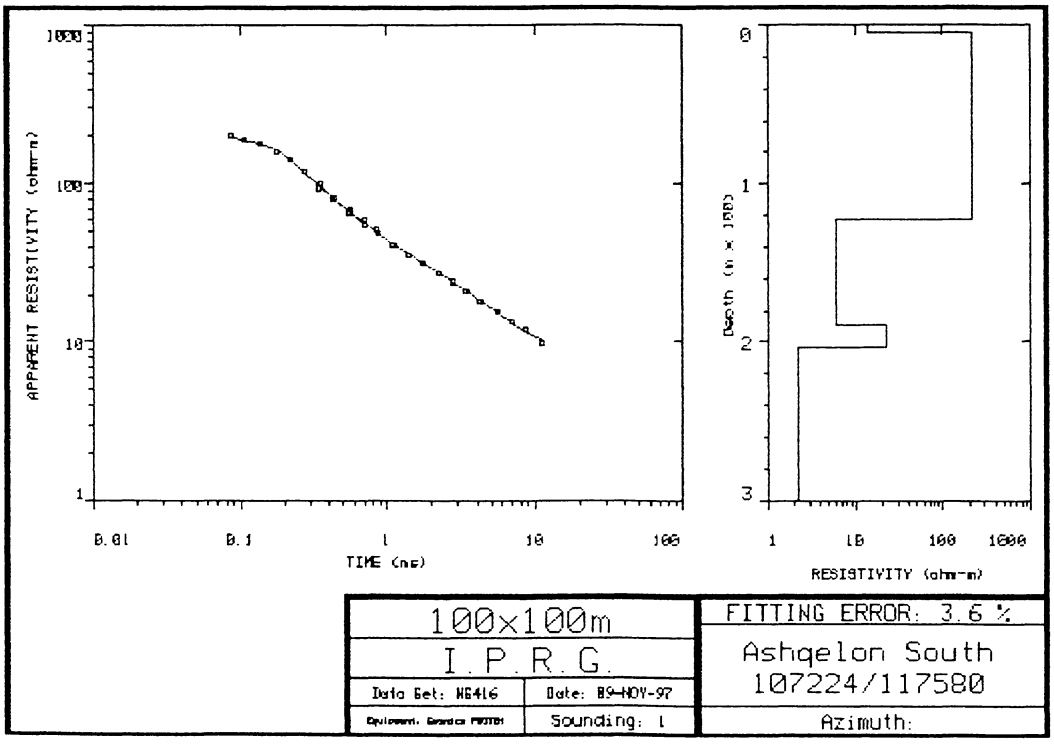
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	4.65761E+01	8.11116E+01
2	1.18211E+00	1.83530E+01
3	5.17579E+01	5.96605E+01
4	2.15150E+00	



150x150m	FITTING ERROR: 1.3 %
I.P.R.G.	Ga'ash 134185/182860
Data Set: HS415 / Date: 10-JUL-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 360

HS416

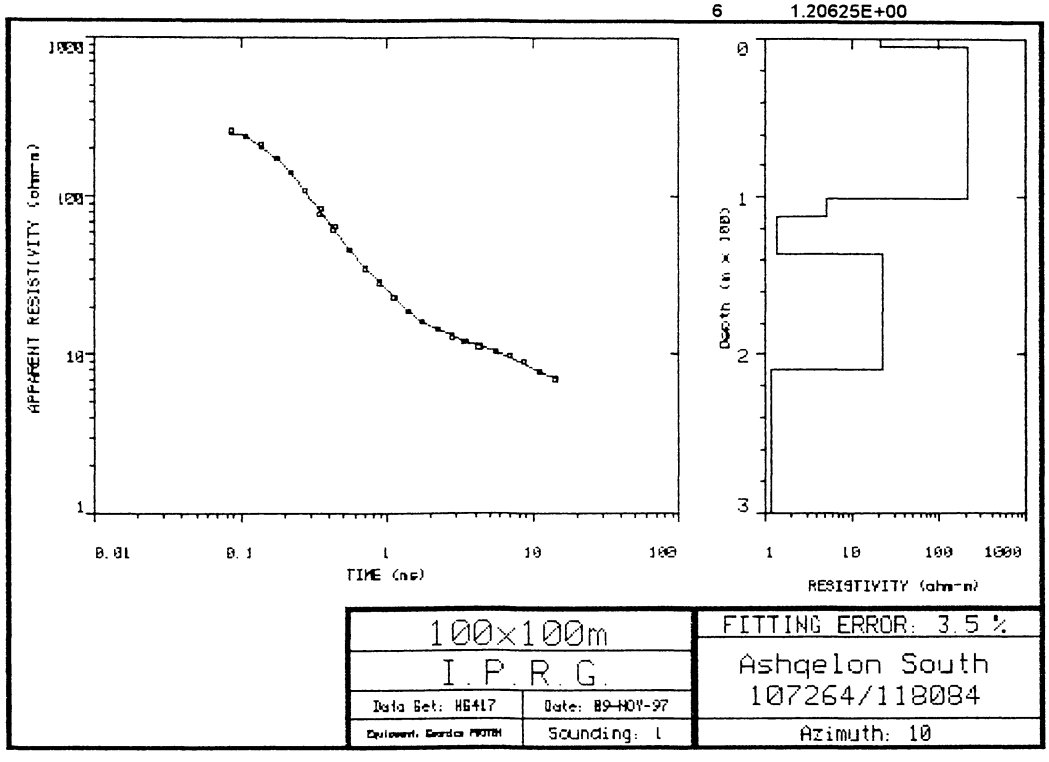
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.43252E+01	4.64315E+00
2	2.15117E+02	1.18120E+02
3	5.99377E+00	6.66738E+01
4	2.19305E+01	1.36255E+01
5	2.23958E+00	



100x100m	FITTING ERROR: 3.6 %
I.P.R.G.	Ashqelon South 107224/117580
Data Set: HS416 / Date: 09-NOV-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth:

HS417

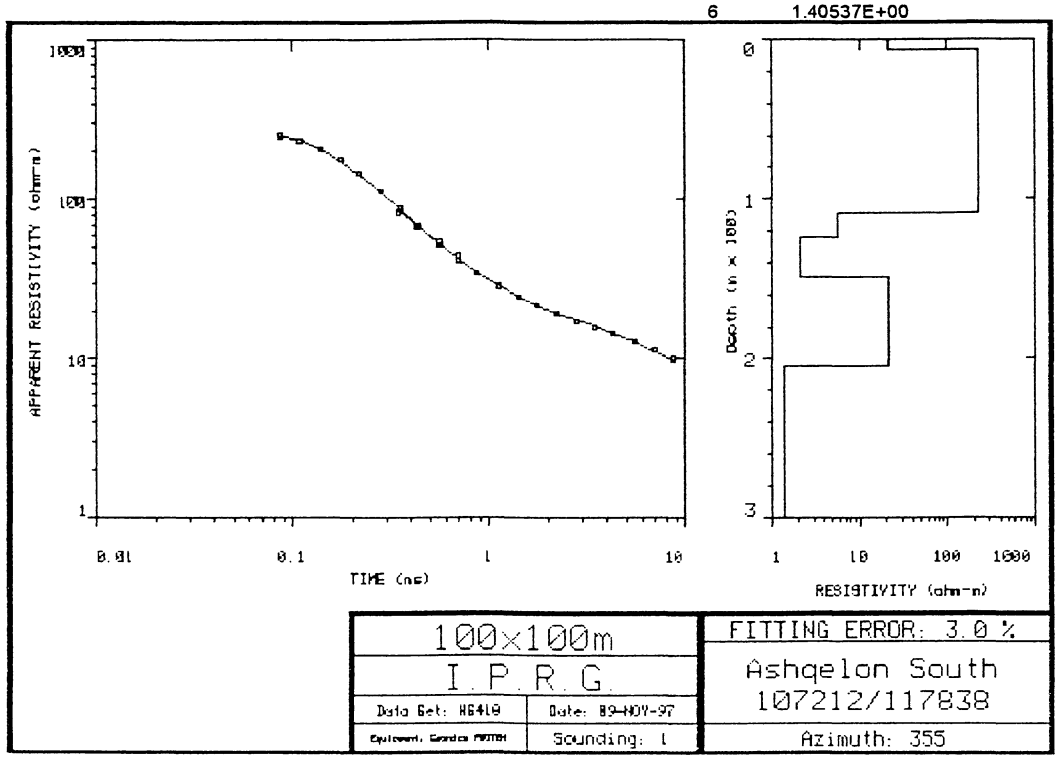
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.12901E+01	5.61135E+00
2	2.16767E+02	9.57248E+01
3	5.23811E+00	1.13963E+01
4	1.35093E+00	2.32880E+01
5	2.27717E+01	7.34418E+01
6	1.20625E+00	



100×100m	FITTING ERROR: 3.5 %
I.P.R.G.	Ashqelon South 107264/118084
Data Set: HS417 / Date: 09-NOV-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 10

HS418

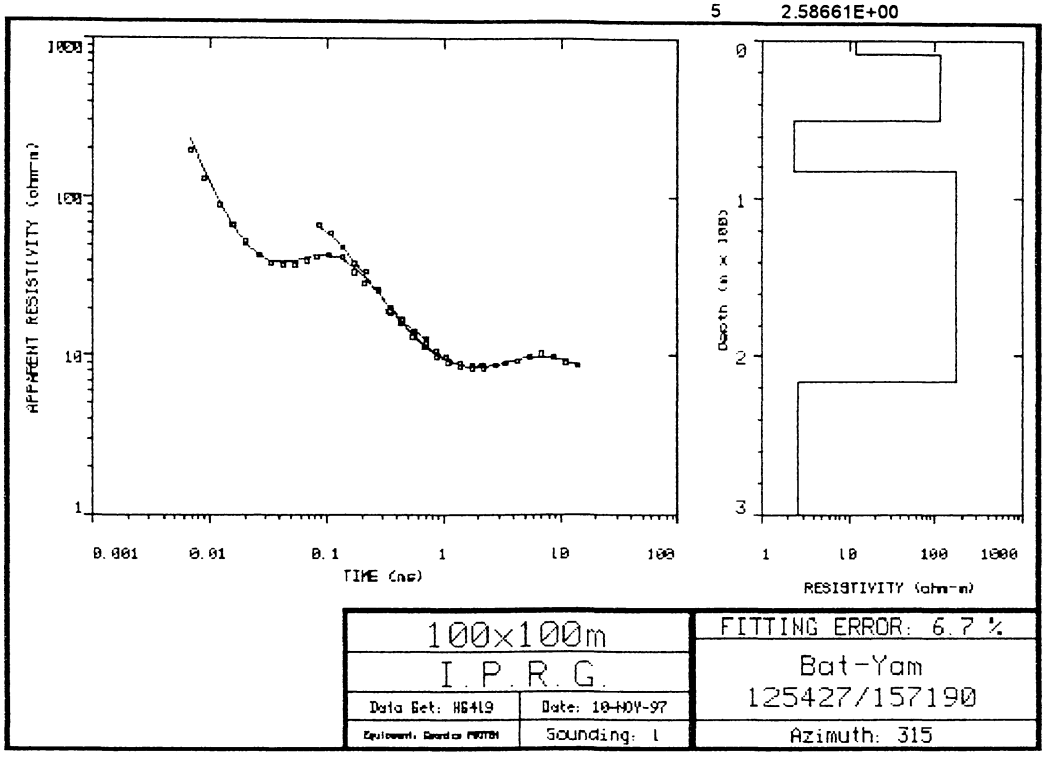
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.17100E+01	5.81977E+00
2	2.32759E+02	1.02334E+02
3	5.84977E+00	1.54817E+01
4	2.09058E+00	2.53019E+01
5	2.17959E+01	5.58675E+01
6	1.40537E+00	



100×100m	FITTING ERROR: 3.0 %
I.P.R.G.	Ashqelon South 107212/117838
Data Set: HS418 / Date: 09-NOV-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 355

HS419

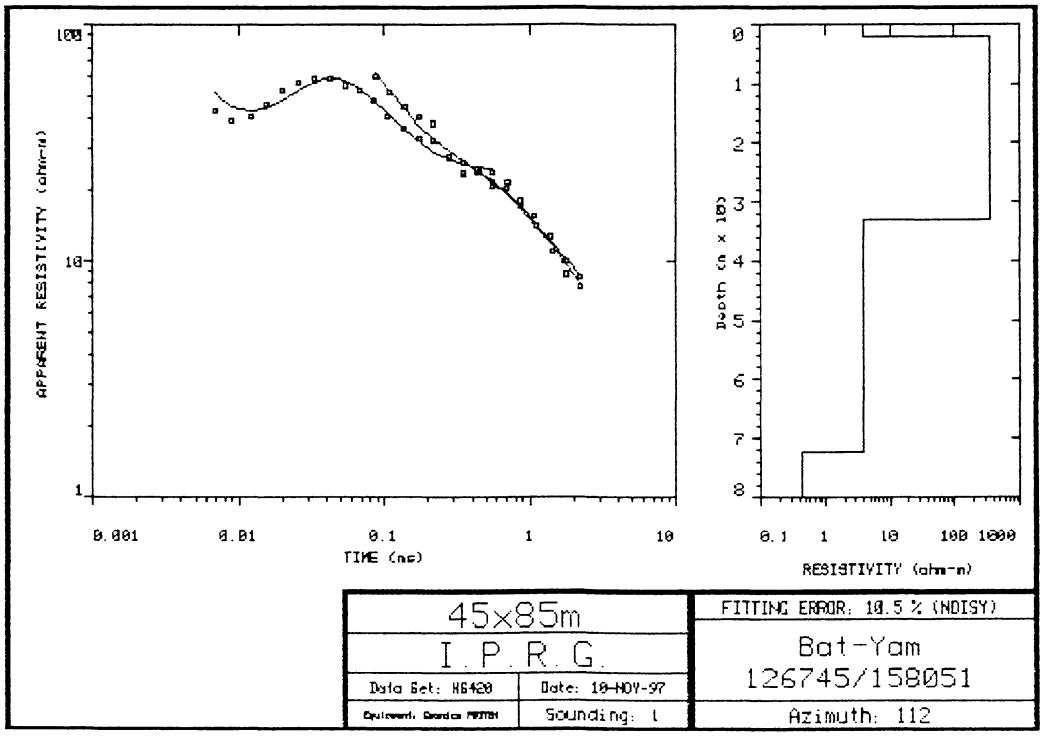
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.18717E+01	8.85909E+00
2	1.12865E+02	4.14266E+01
3	2.33072E+00	3.16975E+01
4	1.74501E+02	1.33613E+02
5	2.58661E+00	



100x100m	FITTING ERROR: 6.7 %
I.P.R.G.	Bat-Yam 125427/157190
Data Set: HS419 / Date: 10-NOV-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 315

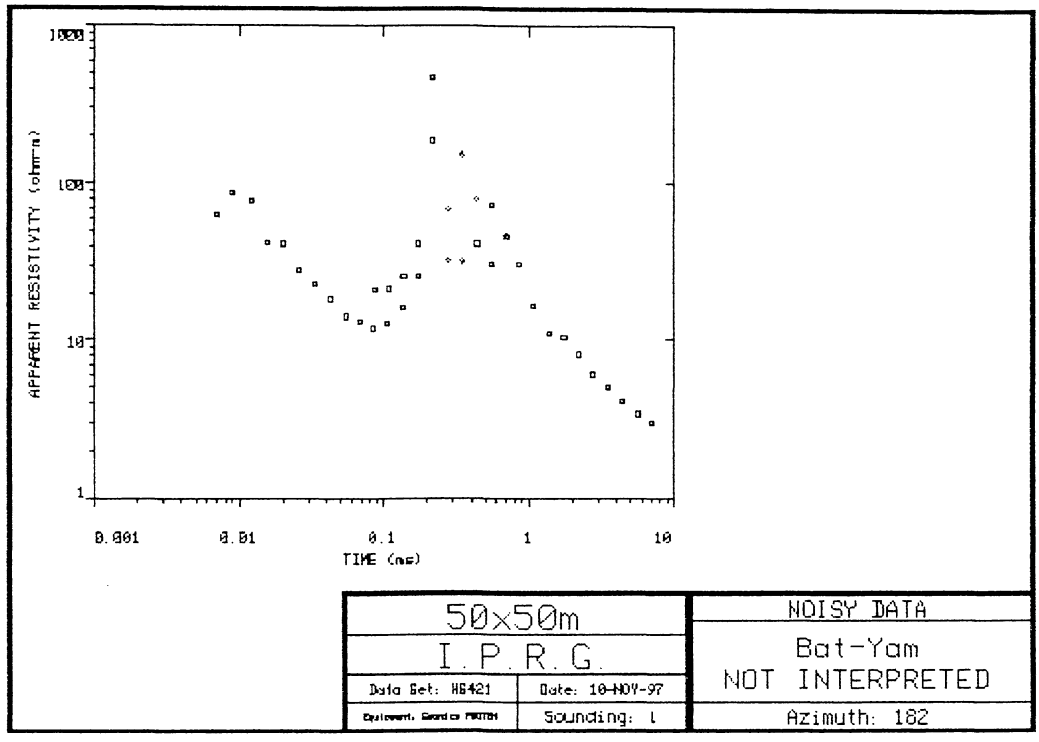
HS420

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	3.85571E+00	1.90014E+00
2	3.51876E+02	3.10795E+01
3	3.85039E+00	3.92513E+01
4	4.43271E-01	



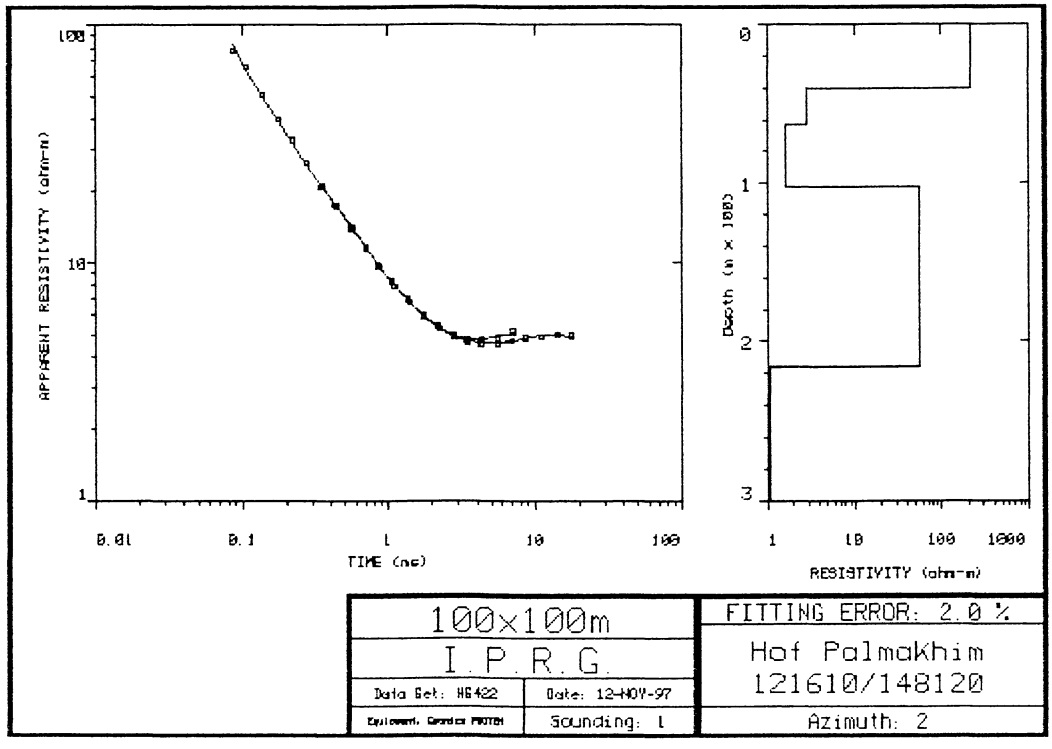
45x85m	FITTING ERROR: 10.5 % (NOISY)
I.P.R.G.	Bat-Yam 126745/158051
Data Set: HS420 / Date: 10-NOV-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 112

HS421



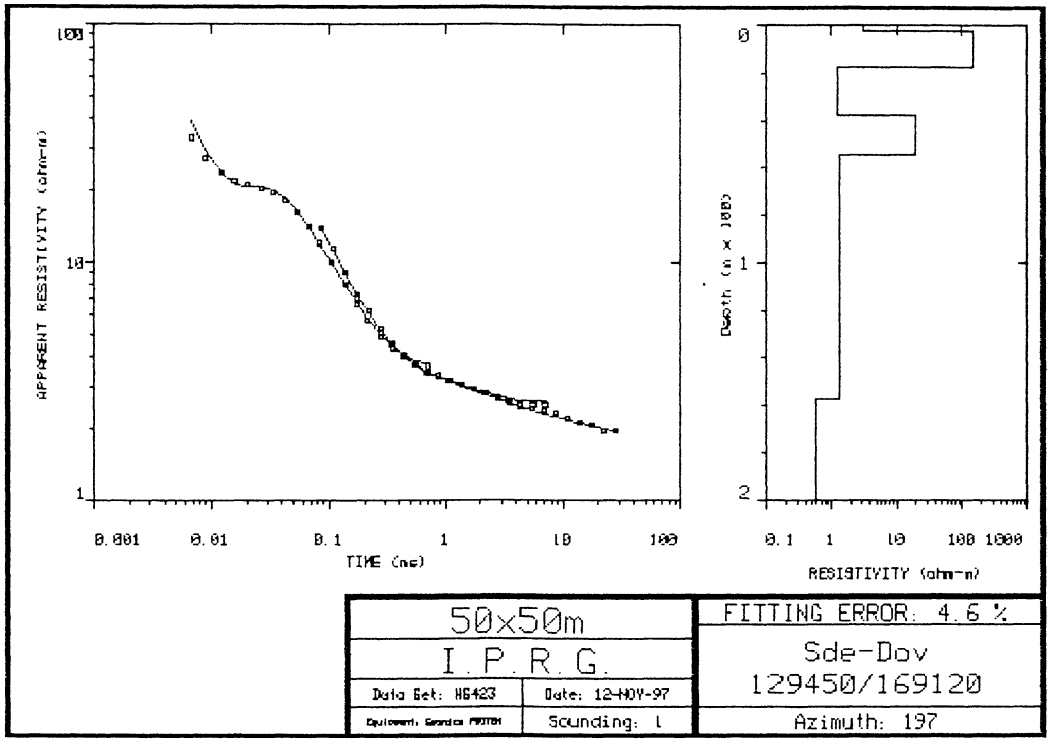
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.21366E+02	4.05833E+01
2	2.83486E+00	2.23101E+01
3	1.59828E+00	3.93184E+01
4	5.42781E+01	1.14099E+02
5	1.03666E+00	

HS422



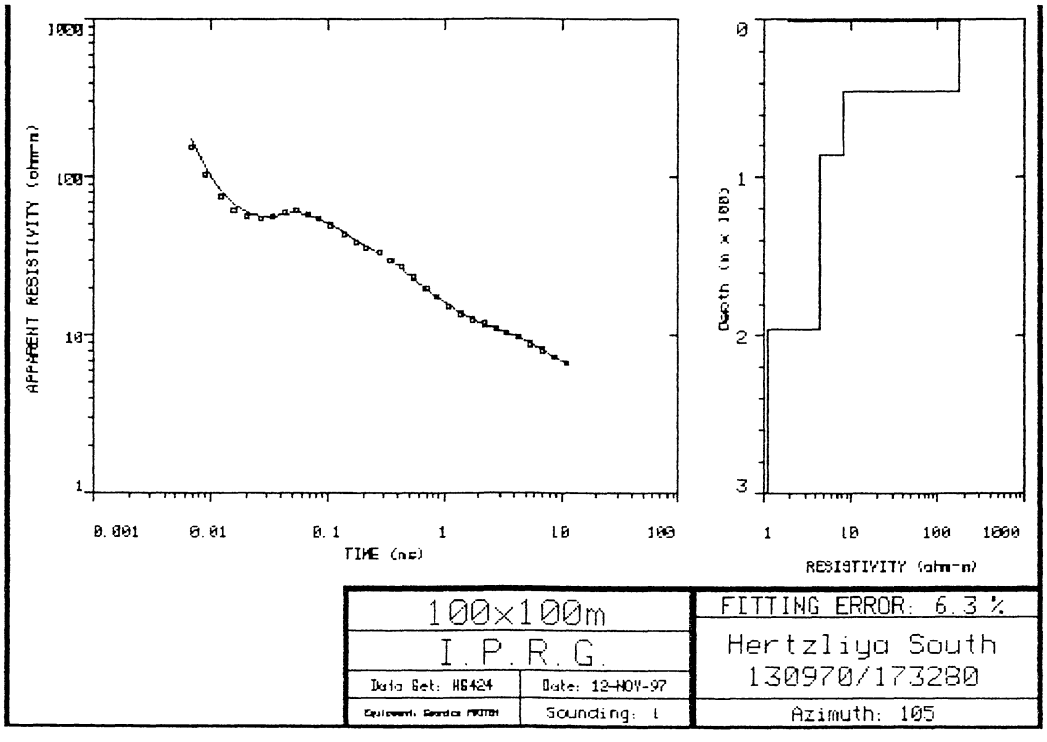
HS423

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	3.08434E+00	2.16602E+00
2	1.45137E+02	1.50995E+01
3	1.29388E+00	2.04147E+01
4	1.88959E+01	1.65824E+01
5	1.35036E+00	1.03192E+02
6	5.93745E-01	



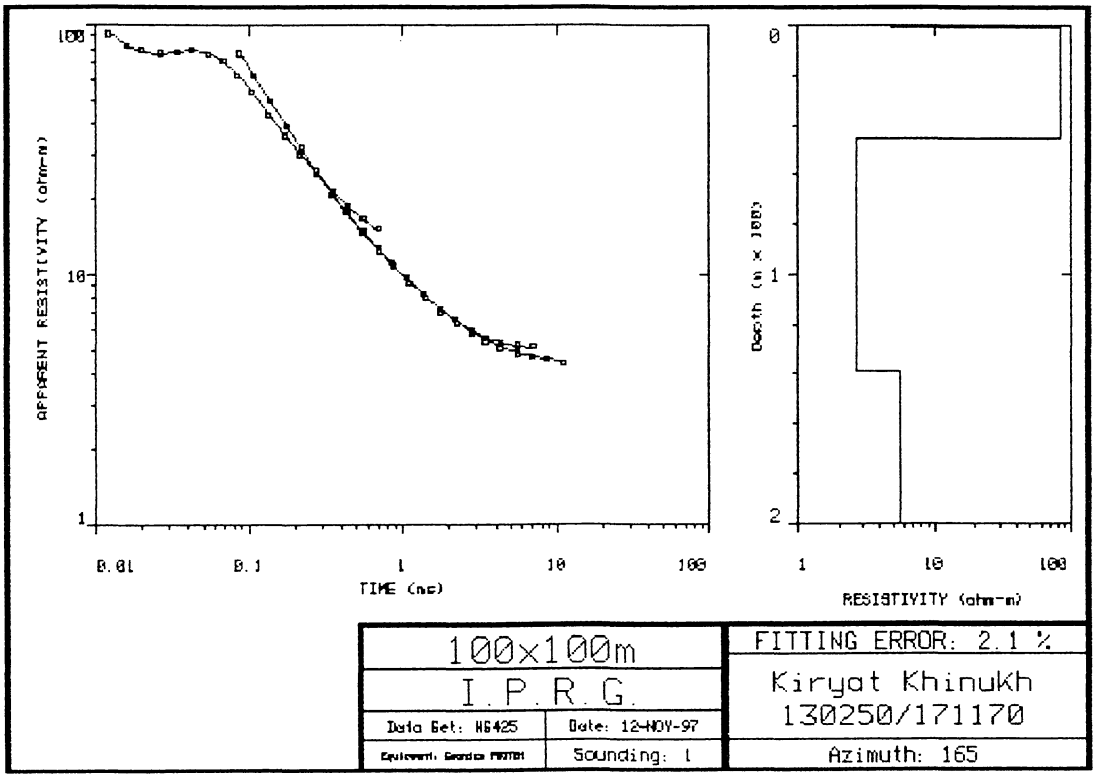
HS424

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.85878E+00	7.72462E-01
2	1.76838E+02	4.43196E+01
3	8.17503E+00	4.03687E+01
4	4.31734E+00	1.10390E+02
5	1.10956E+00	



HS425

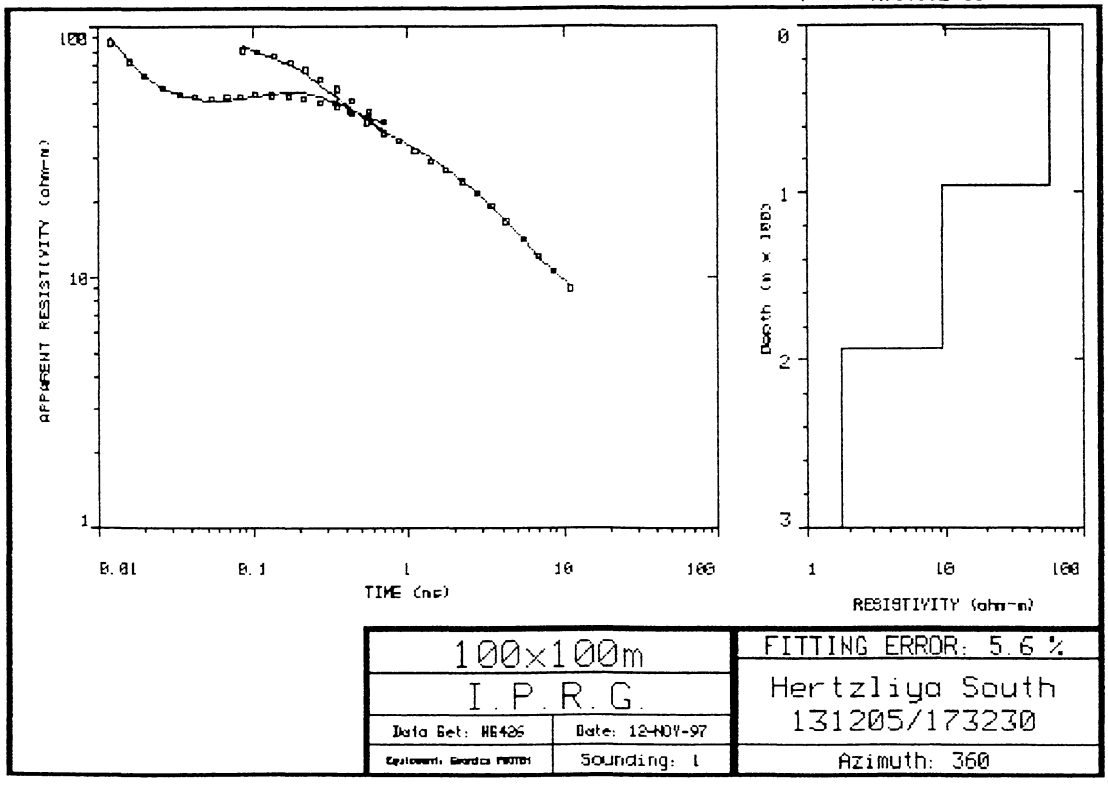
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	4.85516E+00	1.18771E+00
2	8.50160E+01	4.46337E+01
3	2.67036E+00	9.27256E+01
4	5.67609E+00	



100x100m	FITTING ERROR: 2.1 %
I.P.R.G.	Kiryat Khinukh 130250/171170
Data Set: HS425 / Date: 12-NOV-97	
Equipment: Geodica PROTEM / Sounding: 1	Azimuth: 165

HS426

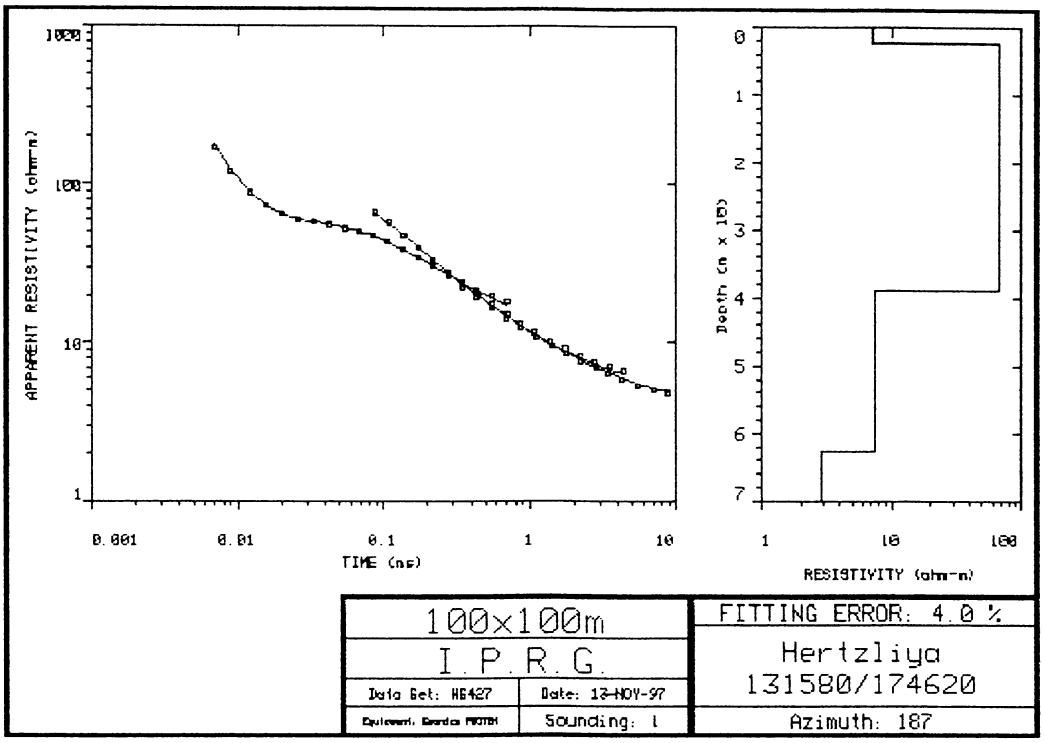
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	9.96773E+00	2.64678E+00
2	5.76847E+01	9.34814E+01
3	9.34851E+00	9.63943E+01
4	1.76156E+00	



100x100m	FITTING ERROR: 5.6 %
I.P.R.G.	Hertzliya South 131205/173230
Data Set: HS426 / Date: 12-NOV-97	
Equipment: Geodica PROTEM / Sounding: 1	Azimuth: 360

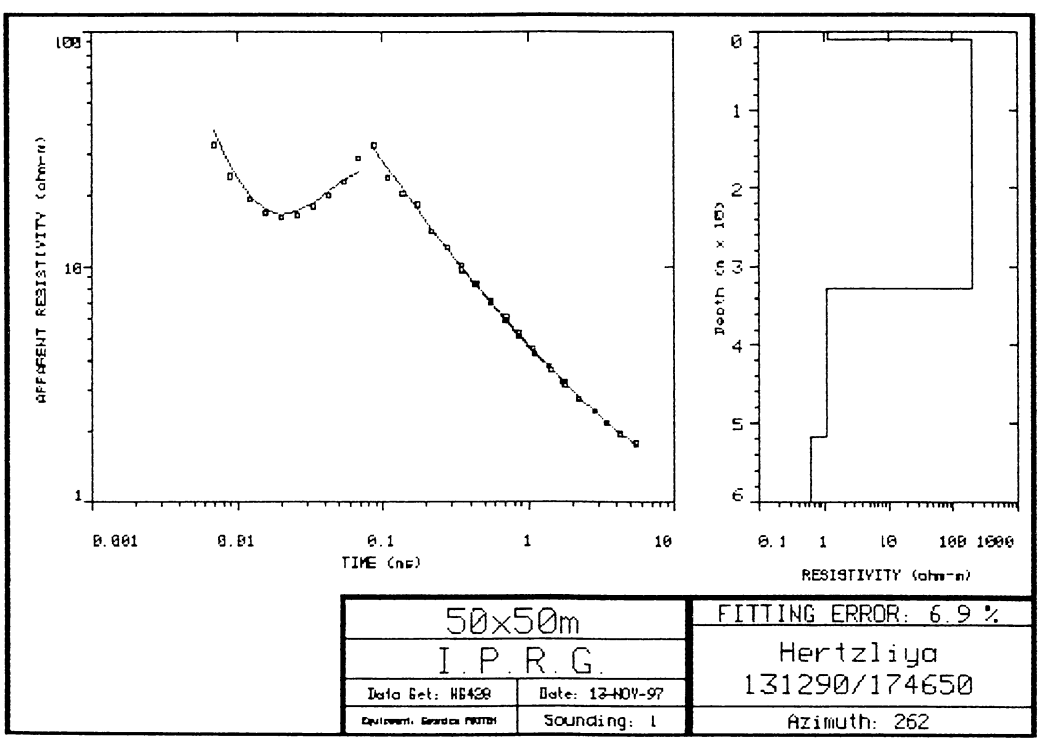
HS427

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	7.13239E+00	2.27454E+00
2	6.81023E+01	3.67385E+01
3	7.30659E+00	2.38545E+01
4	2.88998E+00	



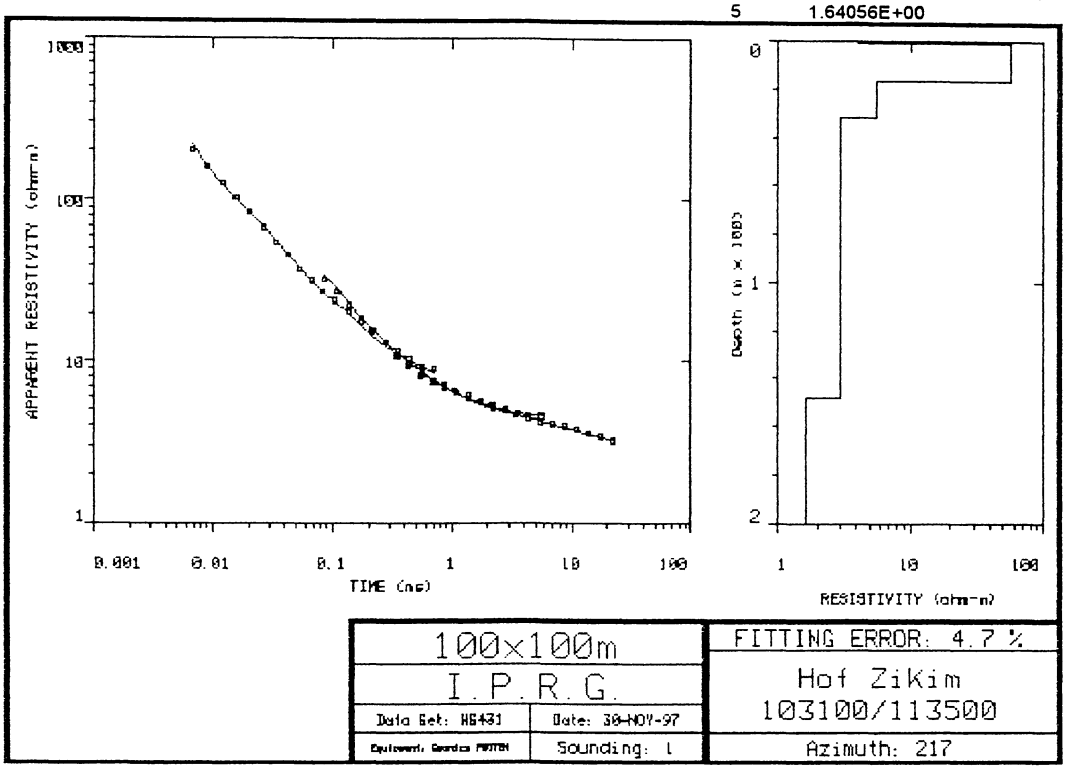
HS428

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.21119E+00	9.28324E-01
2	1.92285E+02	3.19481E+01
3	1.11537E+00	1.87908E+01
4	6.08480E-01	



HS431

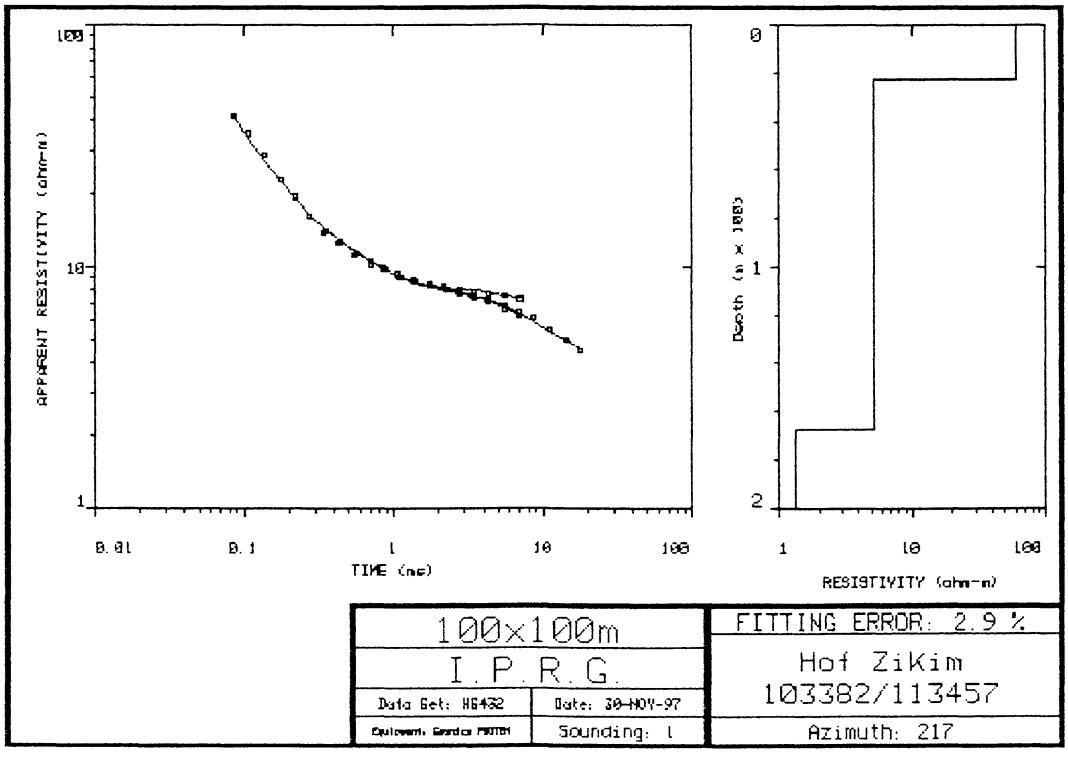
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	4.12189E+00	8.92782E-01
2	5.75365E+01	1.60016E+01
3	5.65466E+00	1.53862E+01
4	2.99914E+00	1.15713E+02
5	1.64056E+00	



100×100m	FITTING ERROR: 4.7 %
I.P.R.G.	Hof ZiKim
Data Set: HS431 / Date: 30-NOV-97	103100/113500
Equipment: Geotron PROTEM / Sounding: 1	Azimuth: 217

HS432

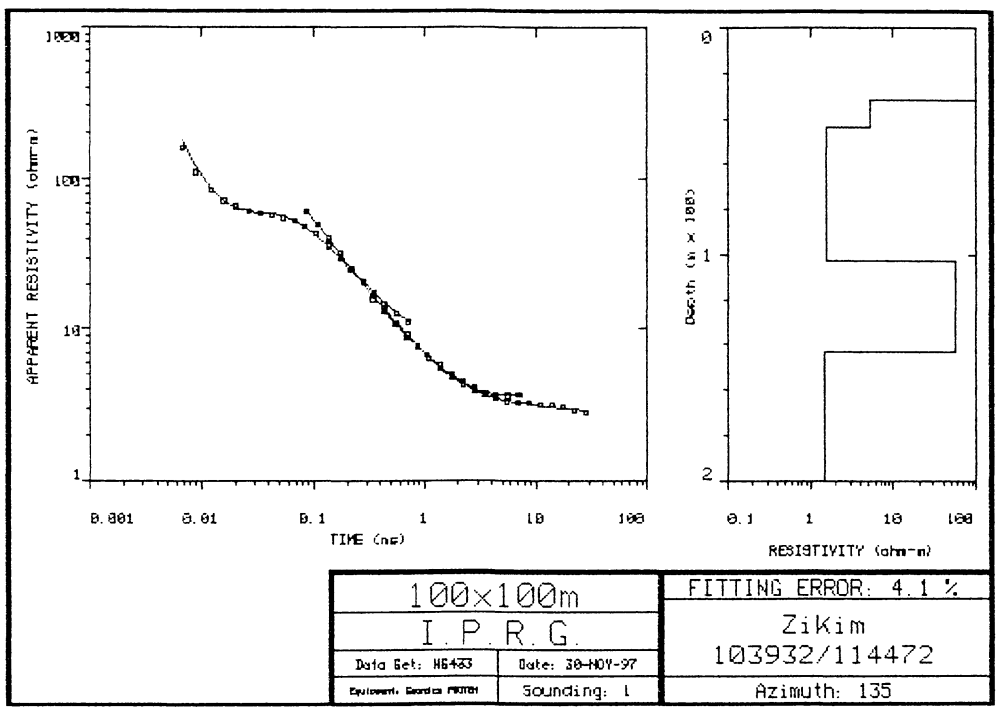
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	5.99732E+01	2.26246E+01
2	5.29819E+00	1.44599E+02
3	1.32479E+00	



100×100m	FITTING ERROR: 2.9 %
I.P.R.G.	Hof ZiKim
Data Set: HS432 / Date: 30-NOV-97	103382/113457
Equipment: Geotron PROTEM / Sounding: 1	Azimuth: 217

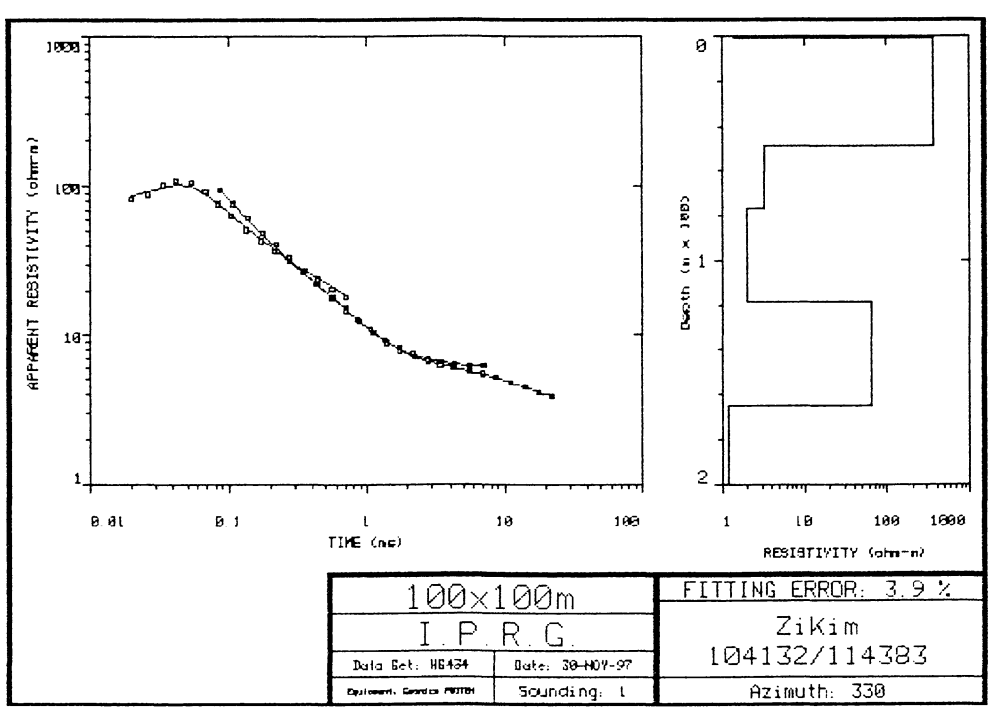
HS433

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	7.74038E-01	2.73395E-01
2	9.87867E+01	3.18602E+01
3	5.18061E+00	1.16833E+01
4	1.52073E+00	5.84405E+01
5	5.49948E+01	4.08144E+01
6	1.48619E+00	



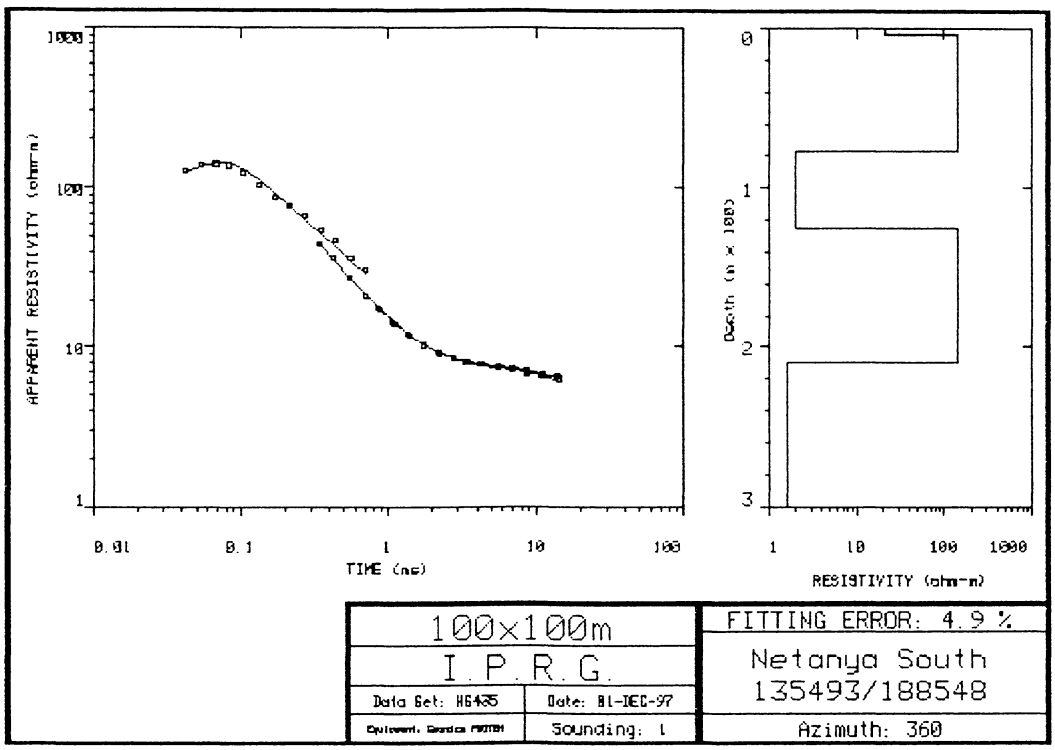
HS434

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.39768E+00	4.27958E-01
2	3.71294E+02	4.85229E+01
3	3.27036E+00	2.78091E+01
4	2.01738E+00	4.15373E+01
5	6.33274E+01	4.63714E+01
6	1.18220E+00	



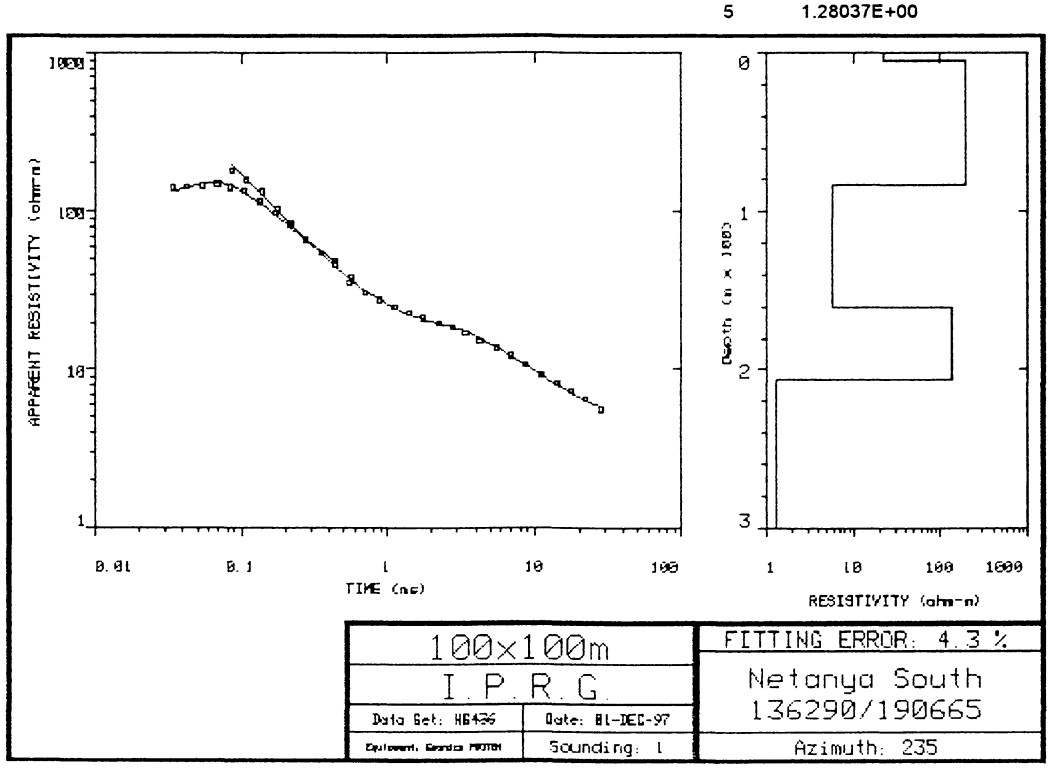
HS435

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.10564E+01	4.33637E+00
2	1.41994E+02	7.20530E+01
3	1.97229E+00	4.88000E+01
4	1.41510E+02	8.37832E+01
5	1.58363E+00	



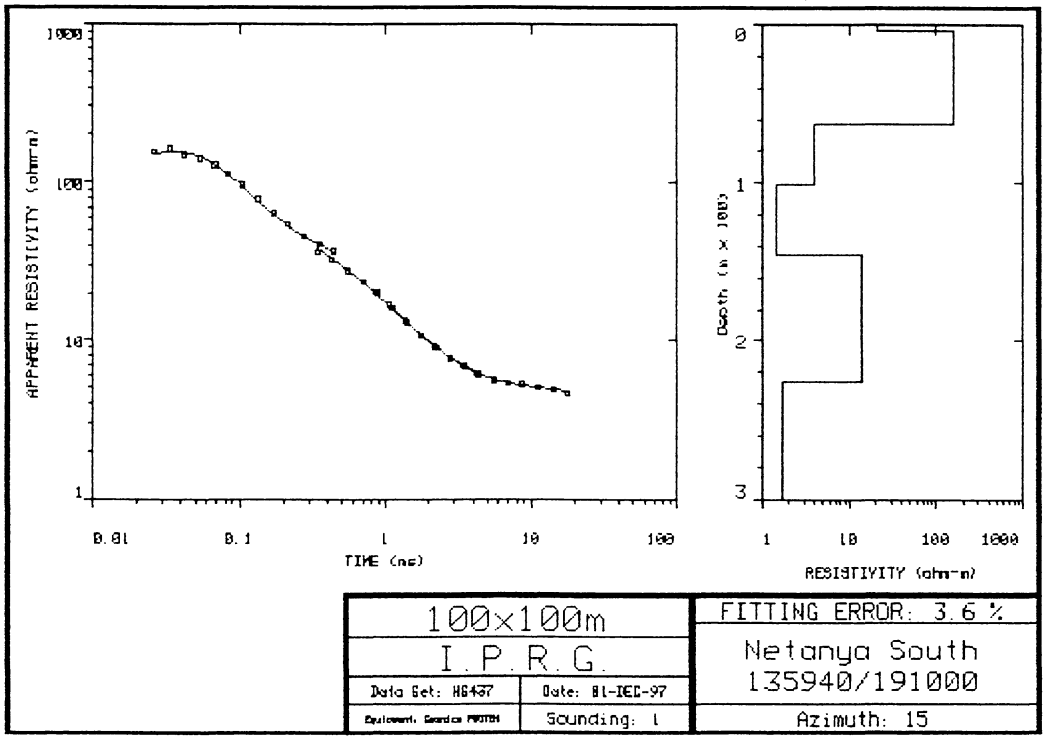
HS436

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.26644E+01	4.44406E+00
2	1.90831E+02	7.87667E+01
3	5.85512E+00	7.70085E+01
4	1.32754E+02	4.53399E+01
5	1.28037E+00	



HS437

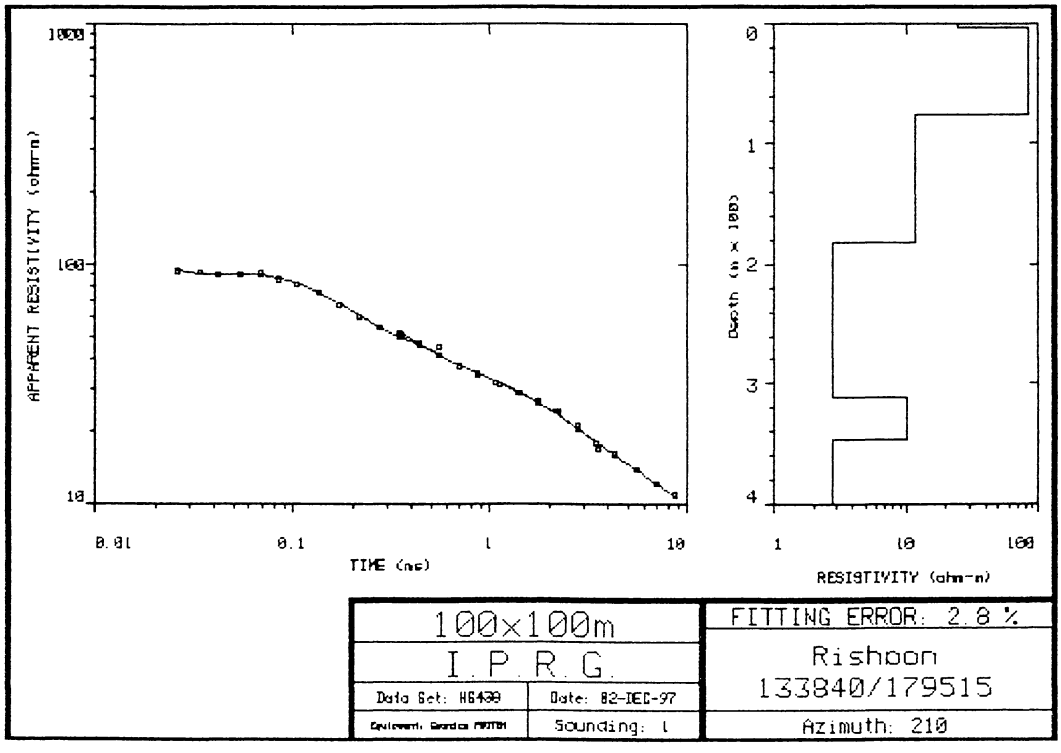
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.11630E+01	3.18415E+00
2	1.60429E+02	5.93724E+01
3	3.98603E+00	3.85853E+01
4	1.41587E+00	4.37804E+01
5	1.40711E+01	8.05600E+01
6	1.72104E+00	



100×100m	FITTING ERROR: 3.6 %
I.P.R.G.	Netanya South
Data Set: HS437 / Date: 01-DEC-97	135940/191000
Equipment: Scintrex IPR12B / Sounding: 1	Azimuth: 15

HS438

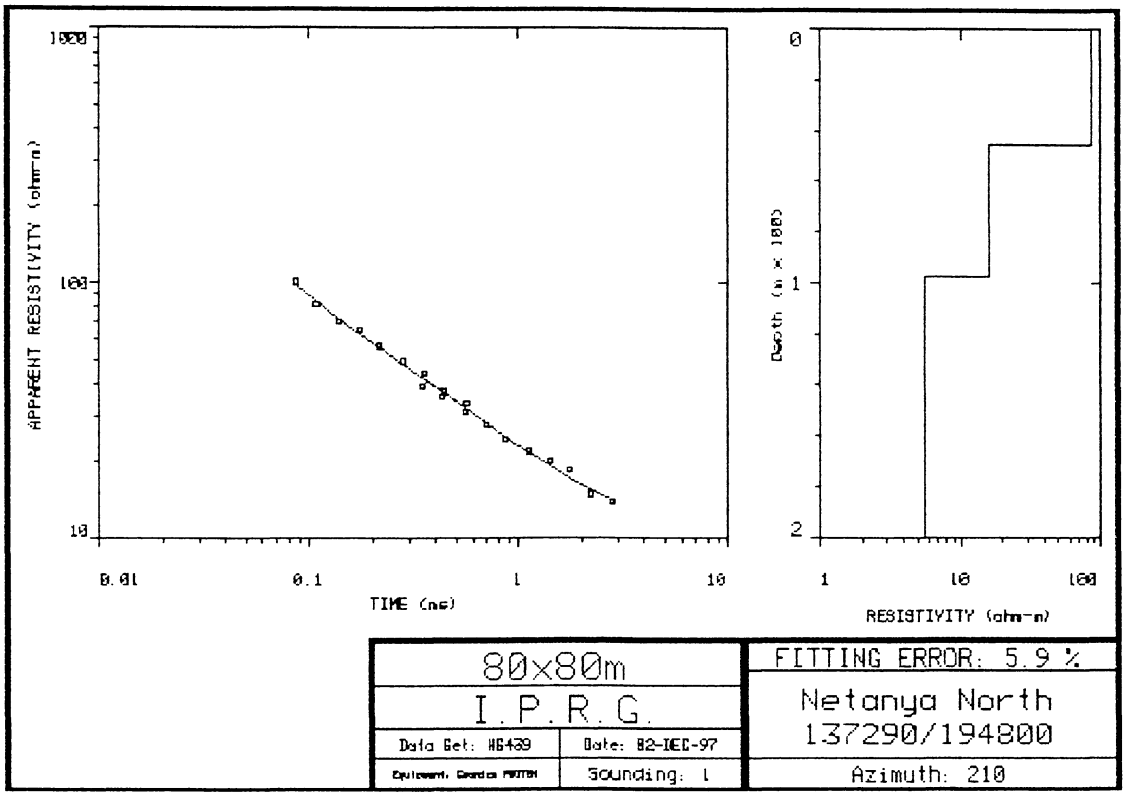
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.41492E+01	3.26617E+00
2	8.34109E+01	7.20735E+01
3	1.17937E+01	1.06874E+02
4	2.80363E+00	1.29319E+02
5	1.03194E+01	3.44456E+01
6	2.73749E+00	



100×100m	FITTING ERROR: 2.8 %
I.P.R.G.	Rishoon
Data Set: HS438 / Date: 02-DEC-97	133840/179515
Equipment: Scintrex IPR12B / Sounding: 1	Azimuth: 210

HS439

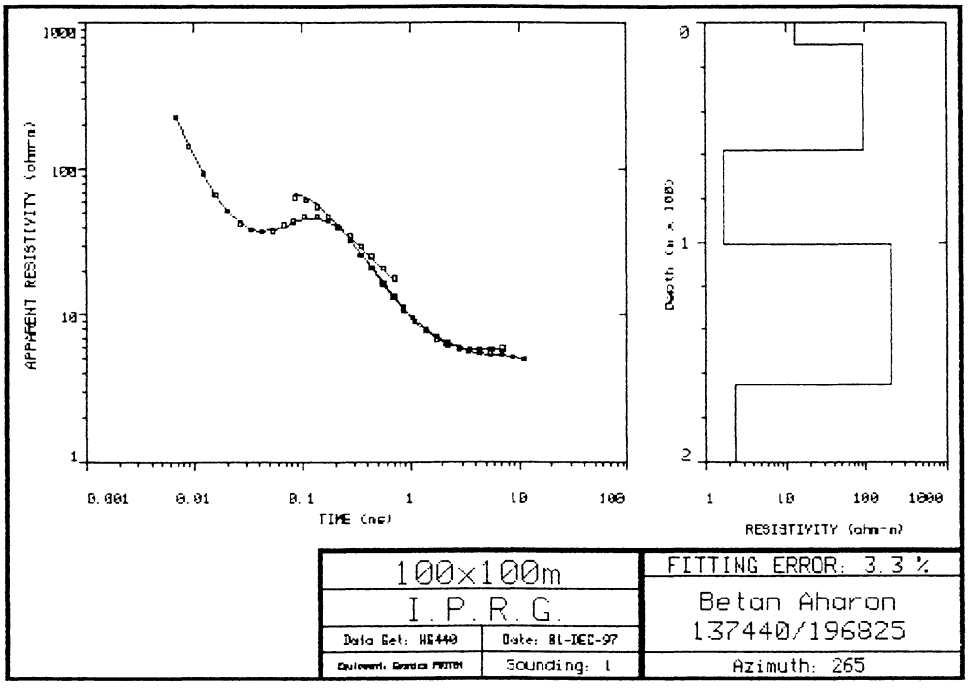
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	8.64953E+01	4.53949E+01
2	1.59432E+01	5.23835E+01
3	5.58337E+00	



80x80m	FITTING ERROR: 5.9 %
I.P.R.G.	Netanya North 137290/194800
Data Set: HS439 / Date: 02-DEC-97	
Equipment: Geotron PROTEM / Sounding: 1	Azimuth: 210

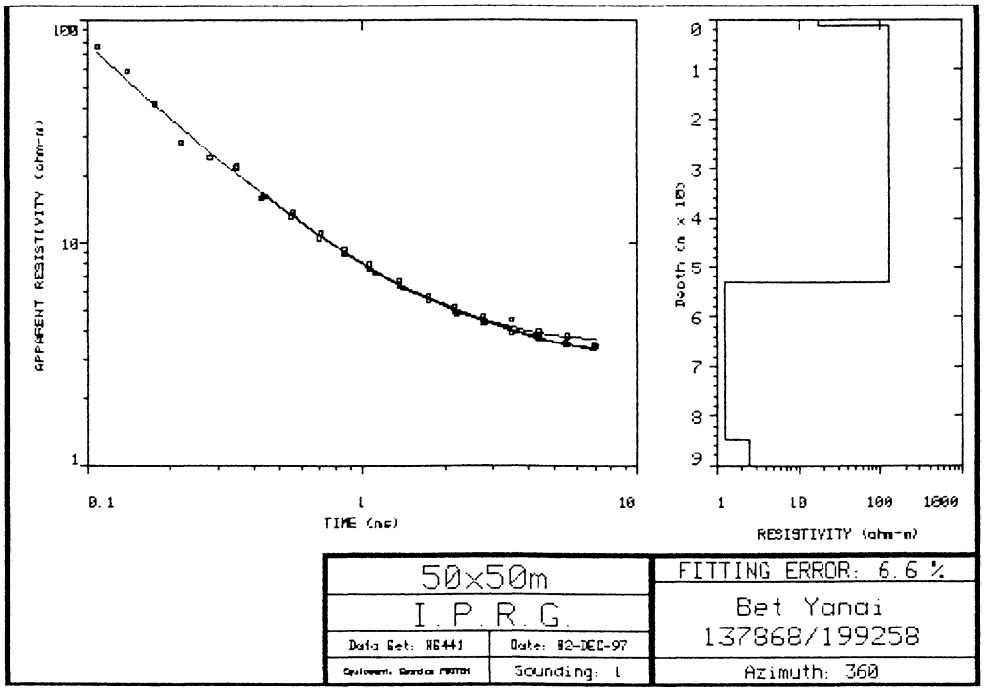
HS440

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.31175E+01	1.01056E+01
2	9.28621E+01	4.79137E+01
3	1.66435E+00	4.25198E+01
4	2.00135E+02	6.40458E+01
5	2.36161E+00	



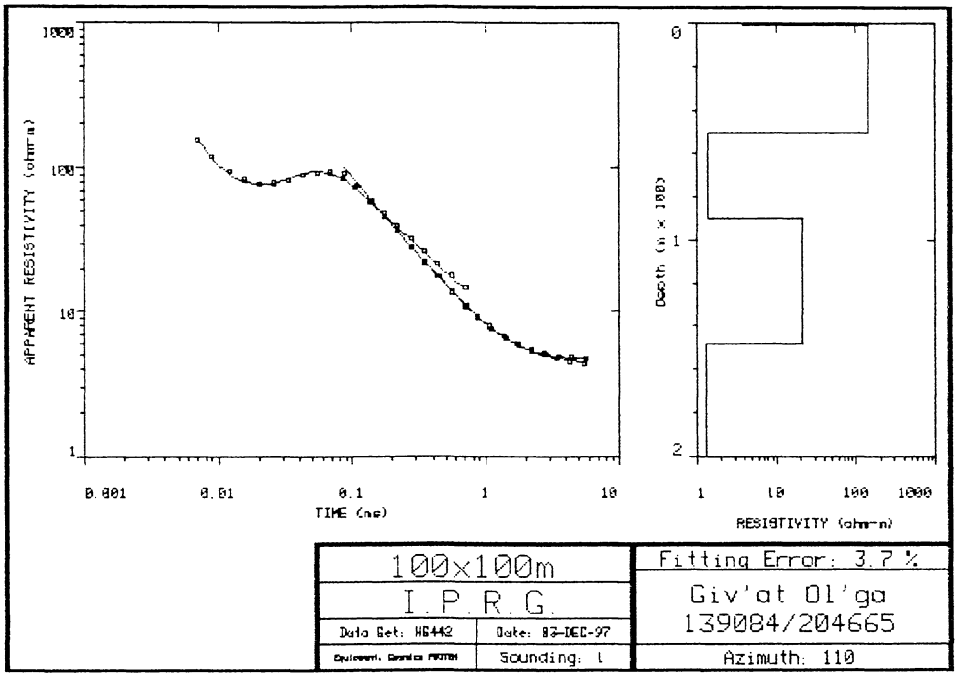
HS441

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.75609E+01	1.31321E+00
2	1.27295E+02	5.15567E+01
3	1.21934E+00	3.20423E+01
4	2.46412E+00	



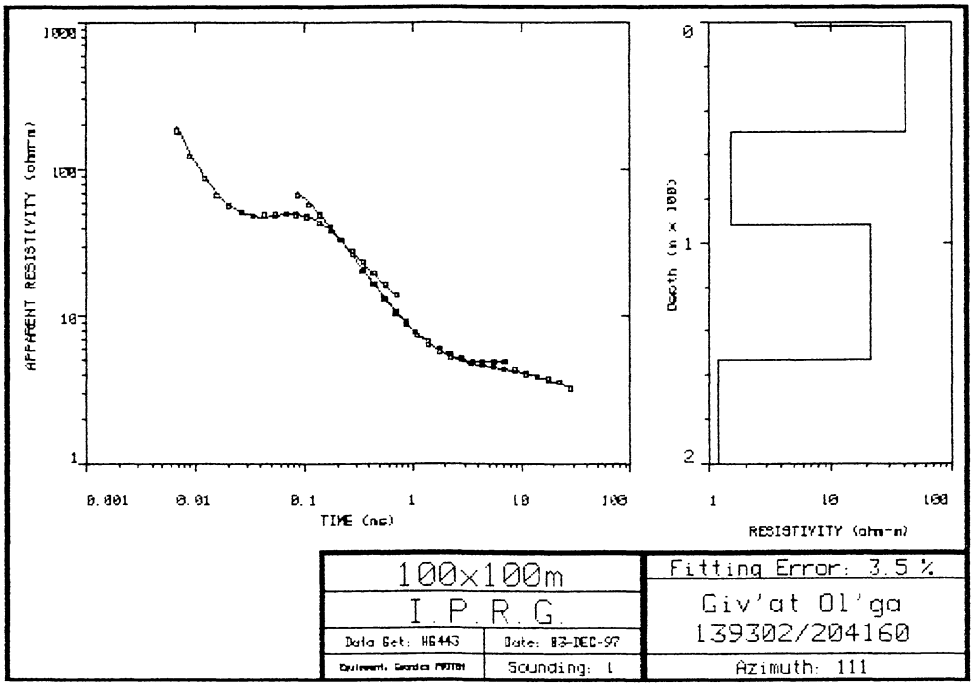
HS442

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	3.67535E+00	1.11230E+00
2	1.43892E+02	4.90445E+01
3	1.40926E+00	3.96091E+01
4	2.11039E+01	5.82953E+01
5	1.29968E+00	



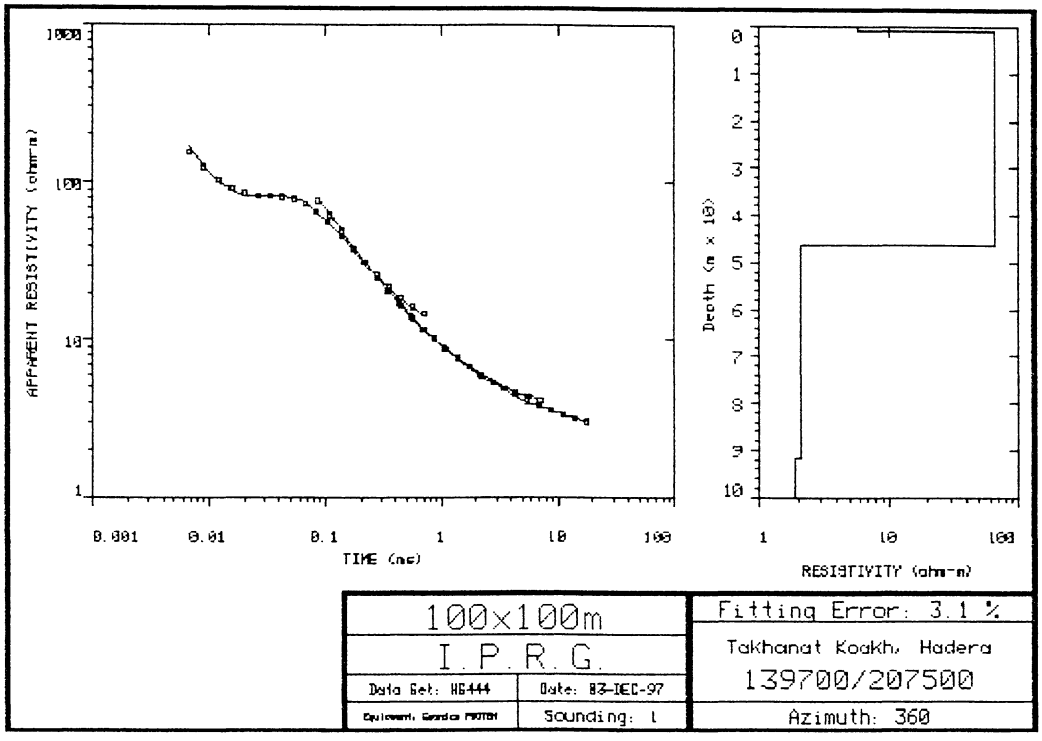
HS443

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	5.19366E+00	1.41472E+00
2	4.18355E+01	4.83820E+01
3	1.52685E+00	4.19010E+01
4	2.11183E+01	6.12487E+01
5	1.21491E+00	



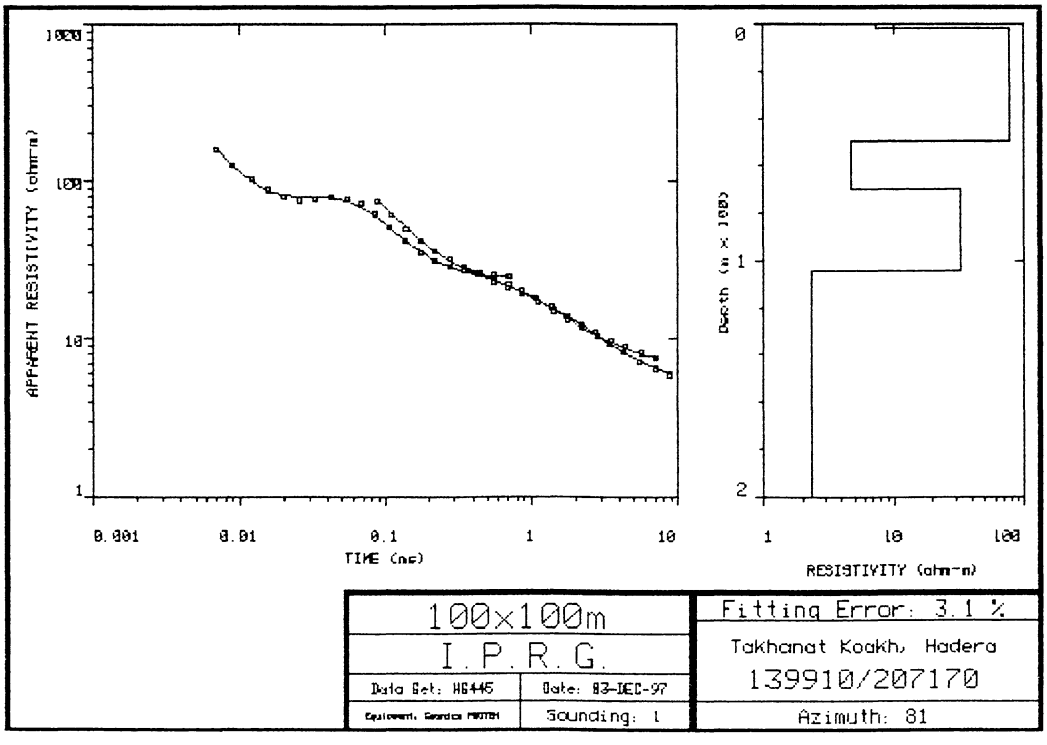
HS444

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	5.79678E+00	9.69460E-01
2	6.49007E+01	4.51821E+01
3	2.09197E+00	4.55041E+01
4	1.85389E+00	



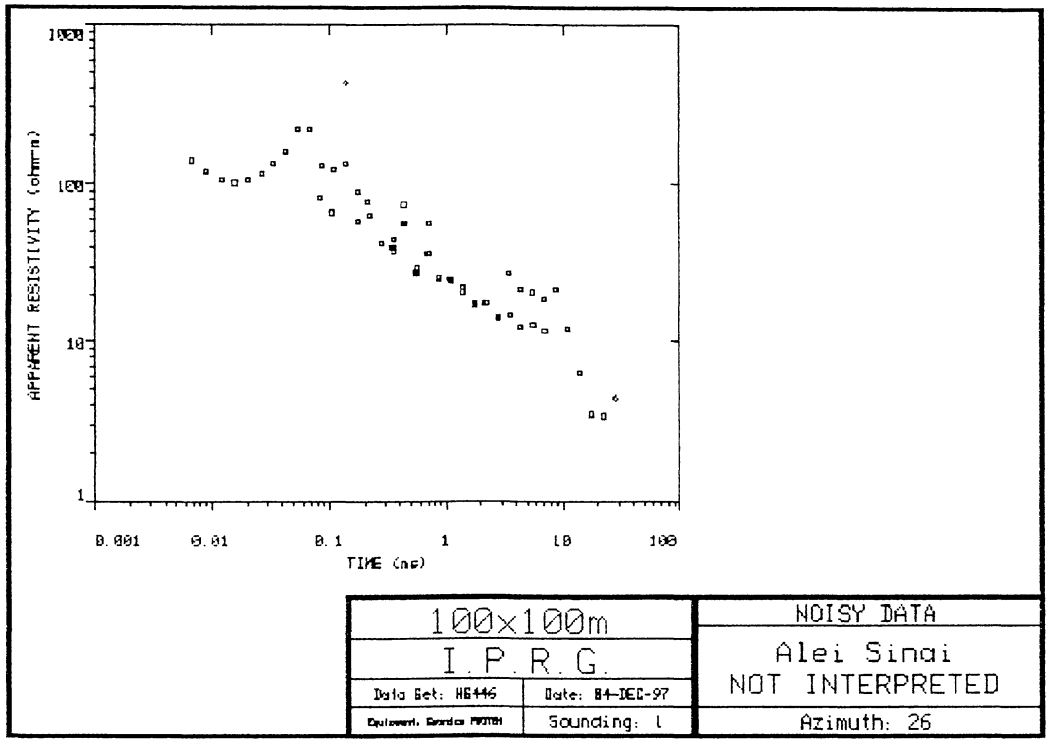
HS445

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	7.29470E+00	1.44114E+00
2	7.71996E+01	4.84426E+01
3	4.70504E+00	1.99618E+01
4	3.21554E+01	3.41332E+01
5	2.29470E+00	



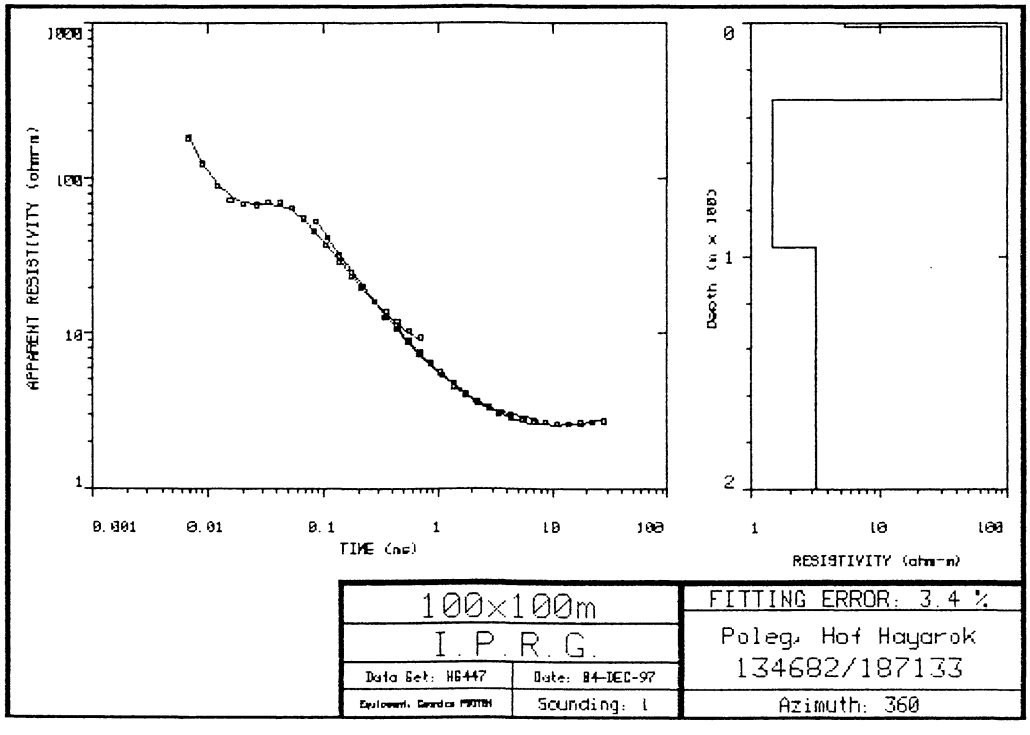
HS446

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	7.29470E+00	1.44114E+00
2	7.71996E+01	4.84426E+01
3	4.70504E+00	1.99618E+01
4	3.21554E+01	3.41332E+01
5	2.29470E+00	



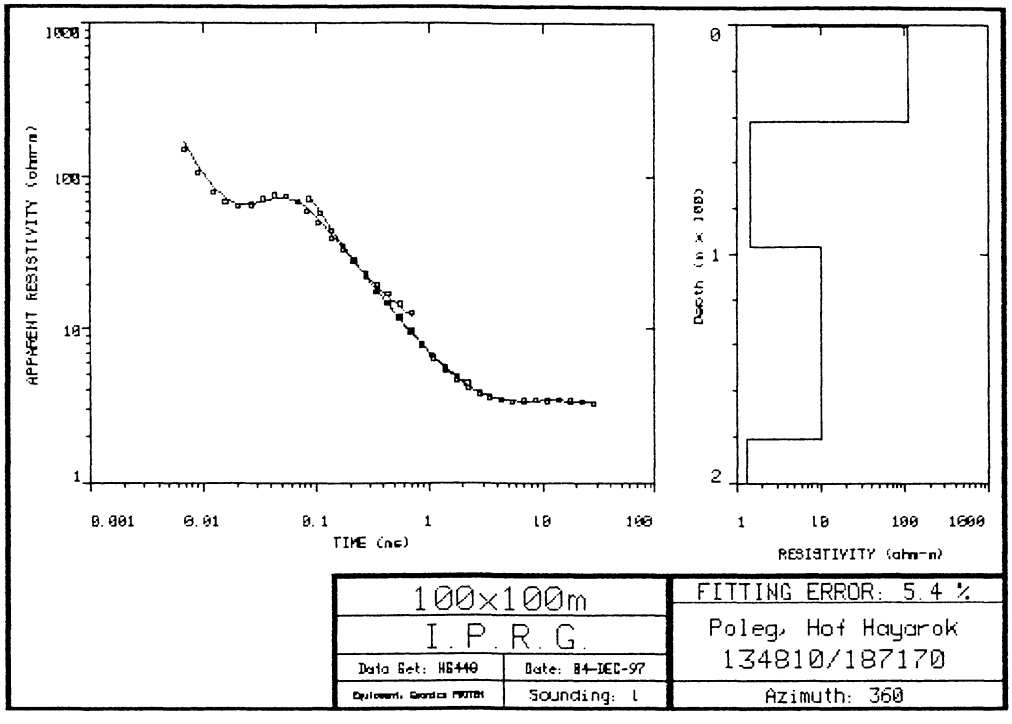
HS447

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	5.50024E+00	1.99738E+00
2	8.94156E+01	3.05984E+01
3	1.45706E+00	6.29501E+01
4	3.17512E+00	



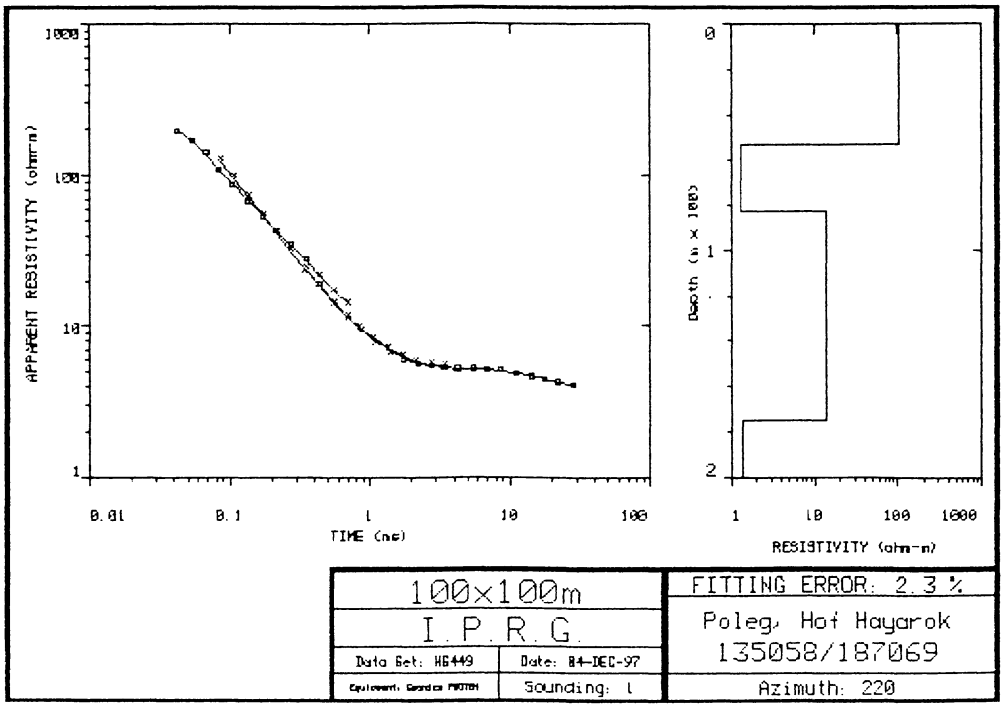
HS448

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.78170E+00	9.47868E-01
2	1.06555E+02	4.10460E+01
3	1.43809E+00	5.48415E+01
4	1.02064E+01	8.37127E+01
5	1.28324E+00	



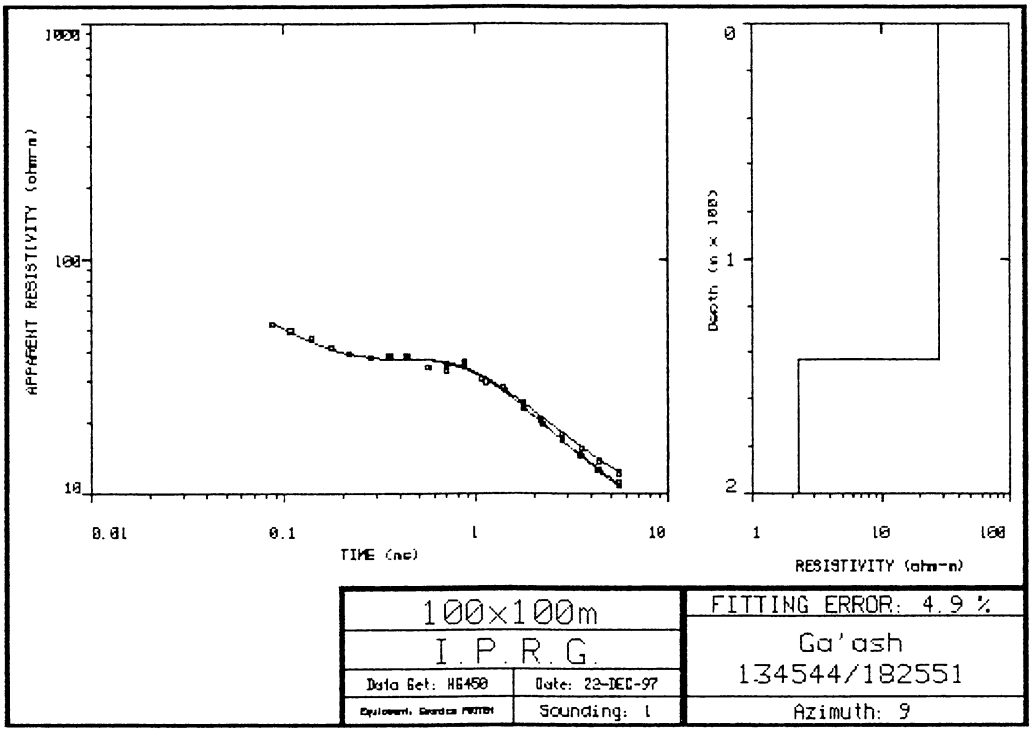
HS449

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.04492E+02	5.30710E+01
2	1.28147E+00	2.90266E+01
3	1.36896E+01	9.32251E+01
4	1.41030E+00	



HS450

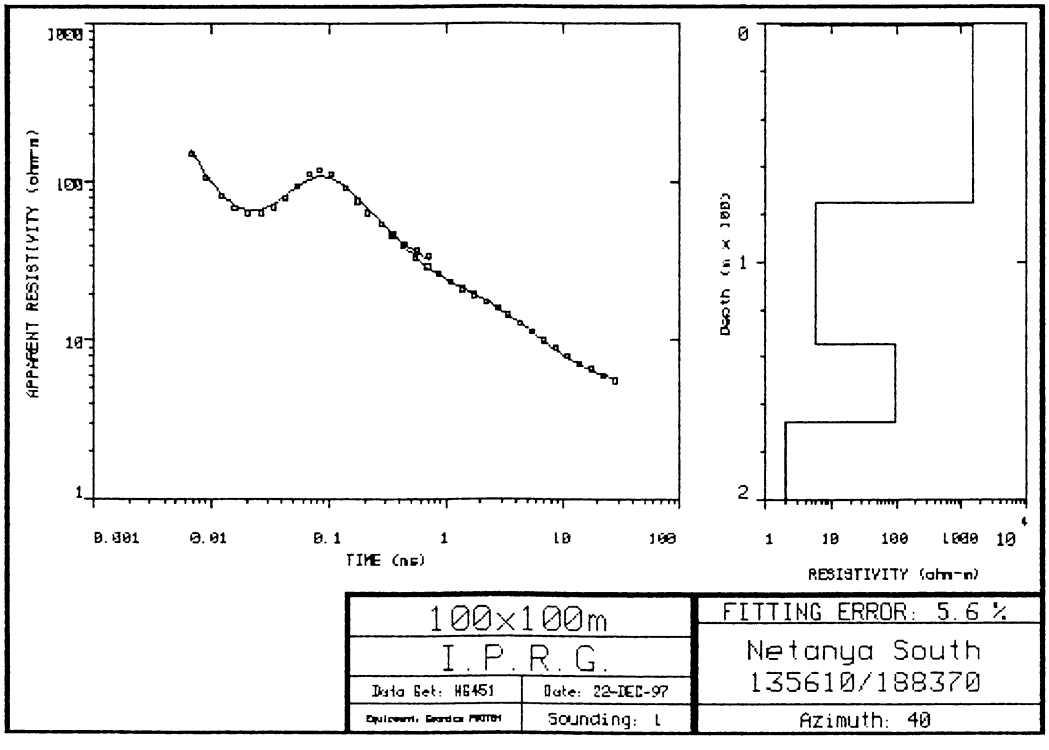
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.76834E+01	1.43315E+02
2	2.21207E+00	



100x100m	FITTING ERROR: 4.9 %
I.P.R.G.	Ga'ash 134544/182551
Data Set: HS450 / Date: 22-DEC-97	
Equipment: Scintrex IPR12 / Sounding: 1	Azimuth: 9

HS451

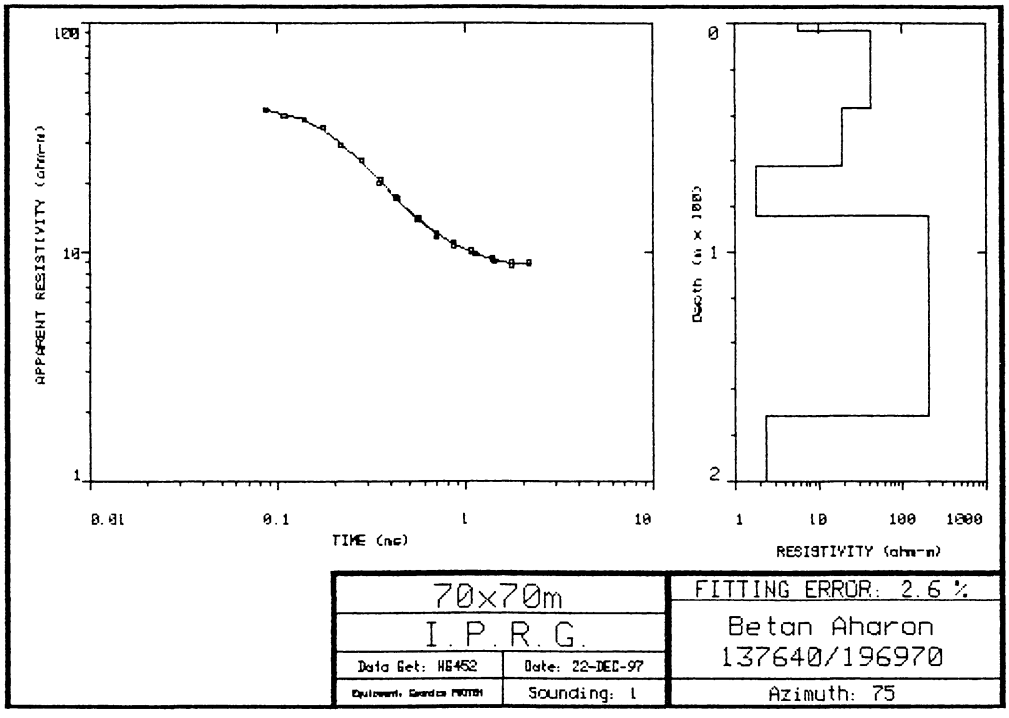
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.75073E+00	7.02936E-01
2	1.47758E+03	7.45058E+01
3	5.67094E+00	5.90490E+01
4	9.57443E+01	3.29213E+01
5	2.04779E+00	



100x100m	FITTING ERROR: 5.6 %
I.P.R.G.	Netanya South 135610/188370
Data Set: HS451 / Date: 22-DEC-97	
Equipment: Scintrex IPR12 / Sounding: 1	Azimuth: 40

HS452

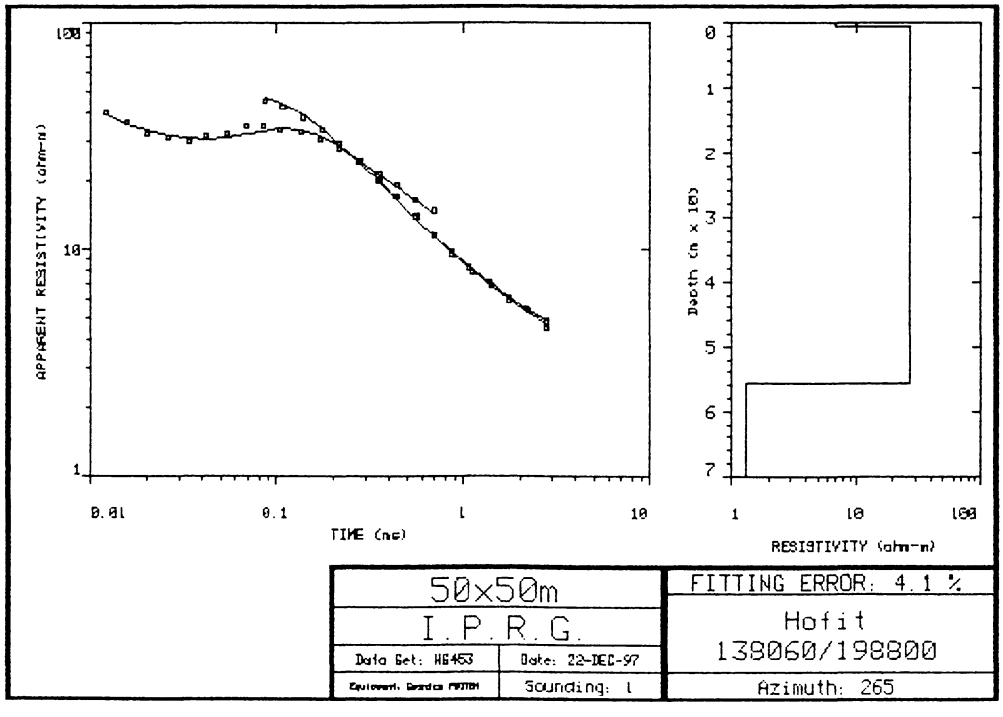
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	5.56327E+00	3.63527E+00
2	4.18475E+01	3.32532E+01
3	1.96658E+01	2.55081E+01
4	1.76283E+00	2.17750E+01
5	2.02780E+02	8.75427E+01
6	2.34299E+00	



70x70m	FITTING ERROR: 2.6 %
I.P.R.G.	Betan Aharon
Data Set: HS452 / Date: 22-DEC-97	137640/196970
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 75

HS453

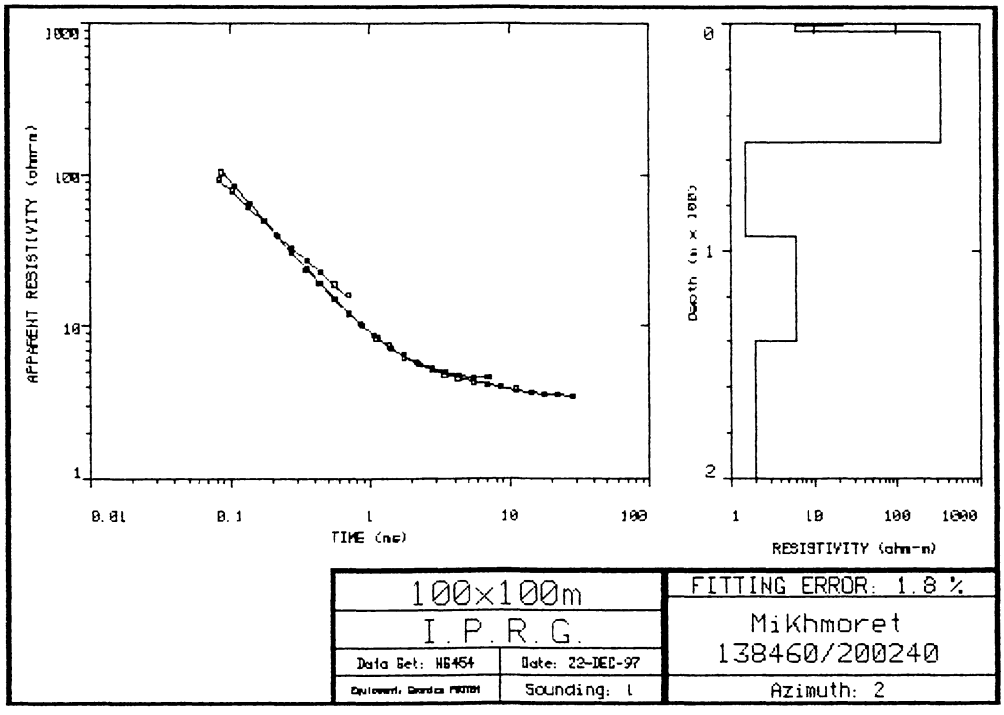
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	6.82954E+00	5.14393E-01
2	2.71150E+01	5.52621E+01
3	1.34874E+00	



50x50m	FITTING ERROR: 4.1 %
I.P.R.G.	Hofit
Data Set: HS453 / Date: 22-DEC-97	138060/198800
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 265

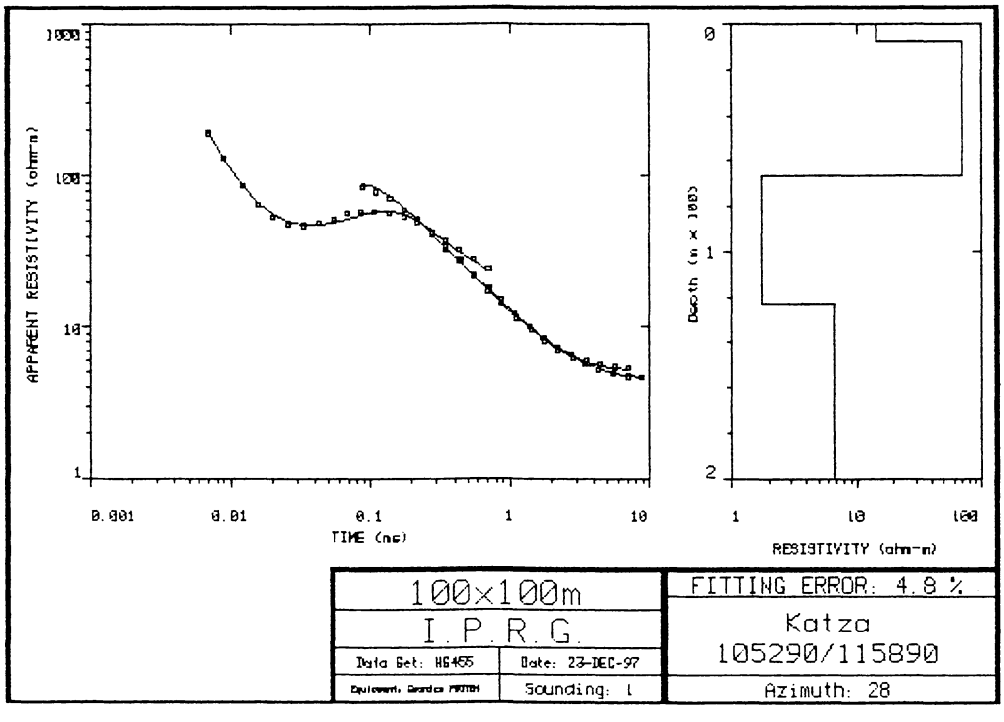
HS454

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.25988E+01	1.19969E+00
2	5.96242E+00	1.73455E+00
3	3.23168E+02	4.91098E+01
4	1.52487E+00	4.12869E+01
5	5.88642E+00	4.65853E+01
6	2.01544E+00	



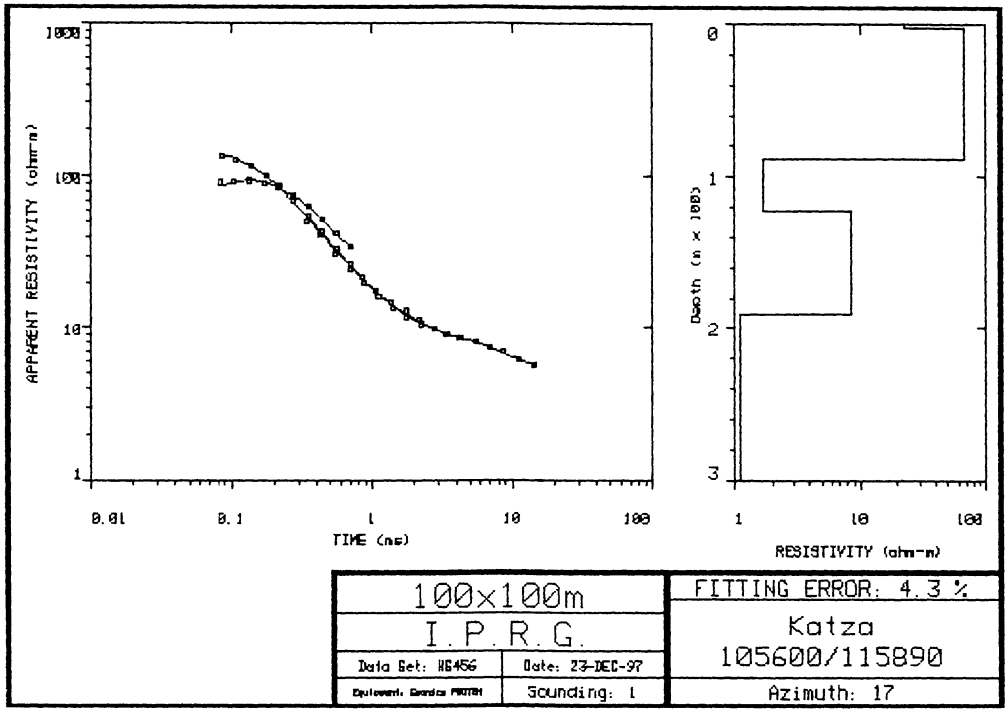
HS455

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.45596E+01	7.22992E+00
2	7.05413E+01	5.92815E+01
3	1.77685E+00	5.62975E+01
4	6.67521E+00	



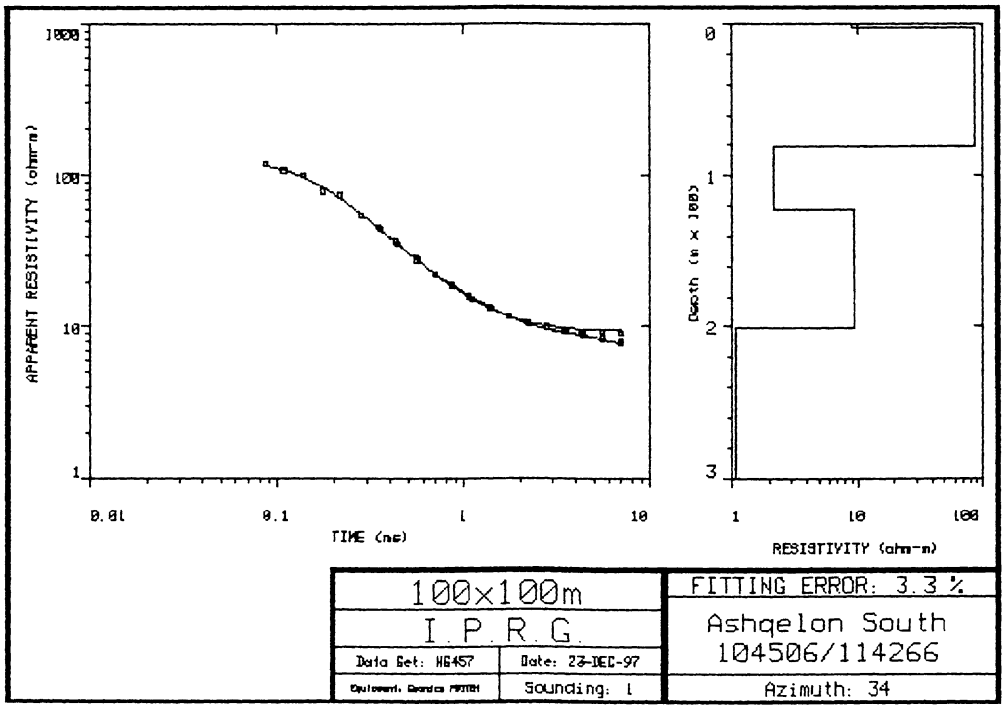
HS456

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	2.26699E+01	2.76116E+00
2	6.74528E+01	8.56585E+01
3	1.67679E+00	3.39889E+01
4	8.60007E+00	6.76808E+01
5	1.09506E+00	



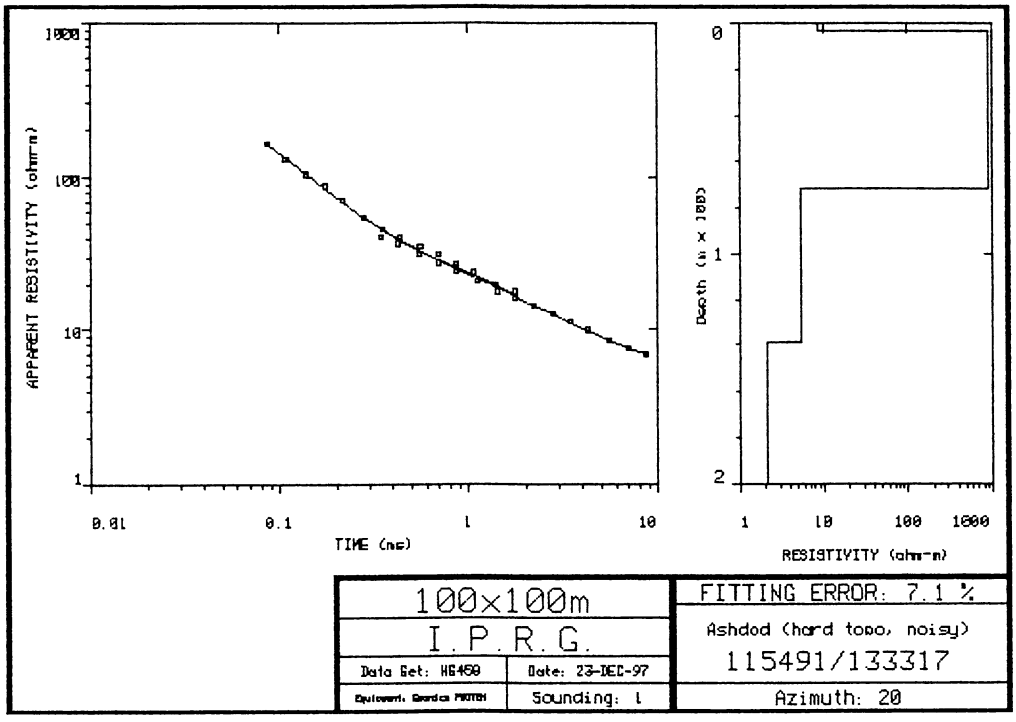
HS457

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	9.22114E+00	3.01817E+00
2	8.59404E+01	7.73445E+01
3	2.17823E+00	4.16011E+01
4	9.35020E+00	7.84687E+01
5	1.06772E+00	



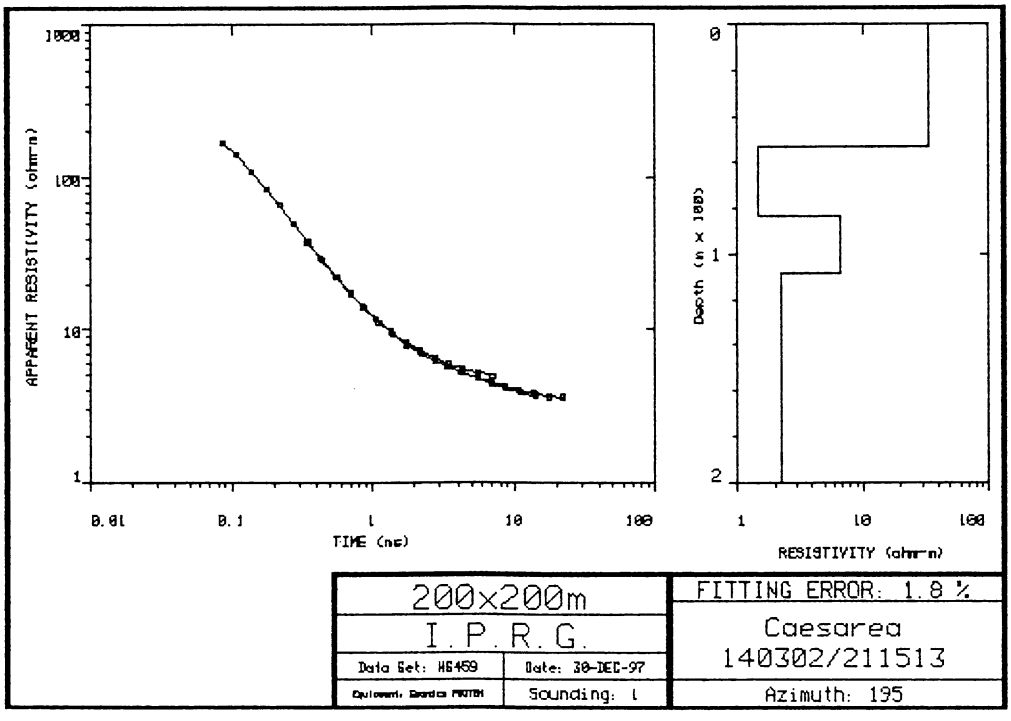
HS458

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	8.70962E+00	2.98225E+00
2	9.13907E+02	6.85408E+01
3	5.29491E+00	6.74183E+01
4	2.08785E+00	



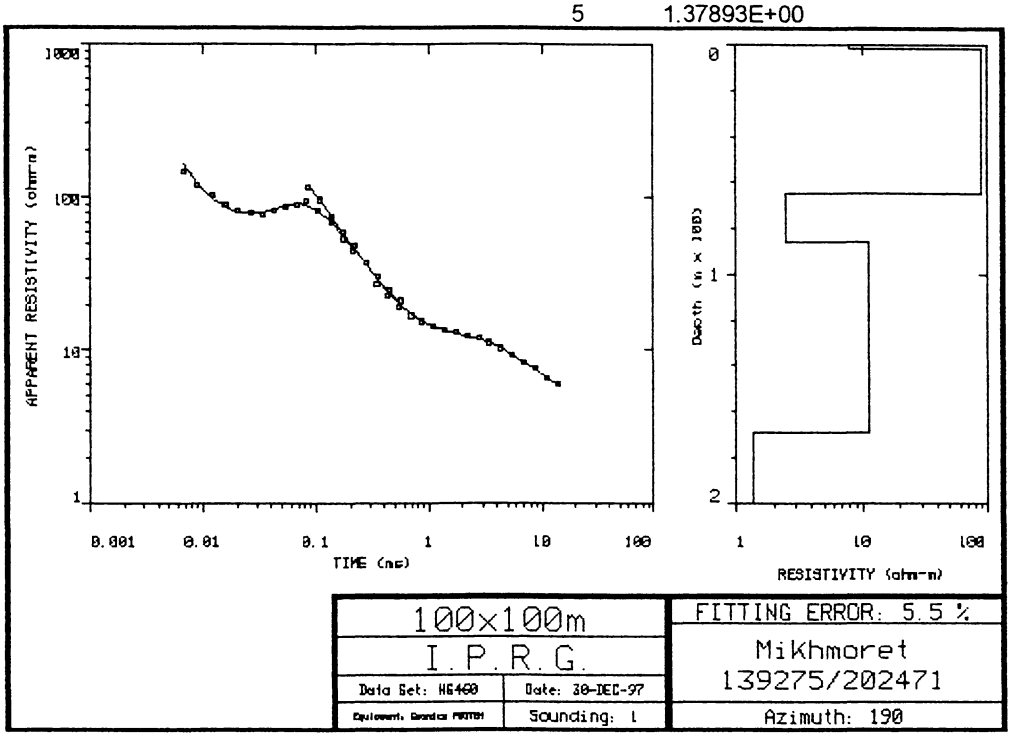
HS459

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	3.36901E+01	5.32654E+01
2	1.47243E+00	3.00452E+01
3	6.65926E+00	2.50152E+01
4	2.25591E+00	



HS460

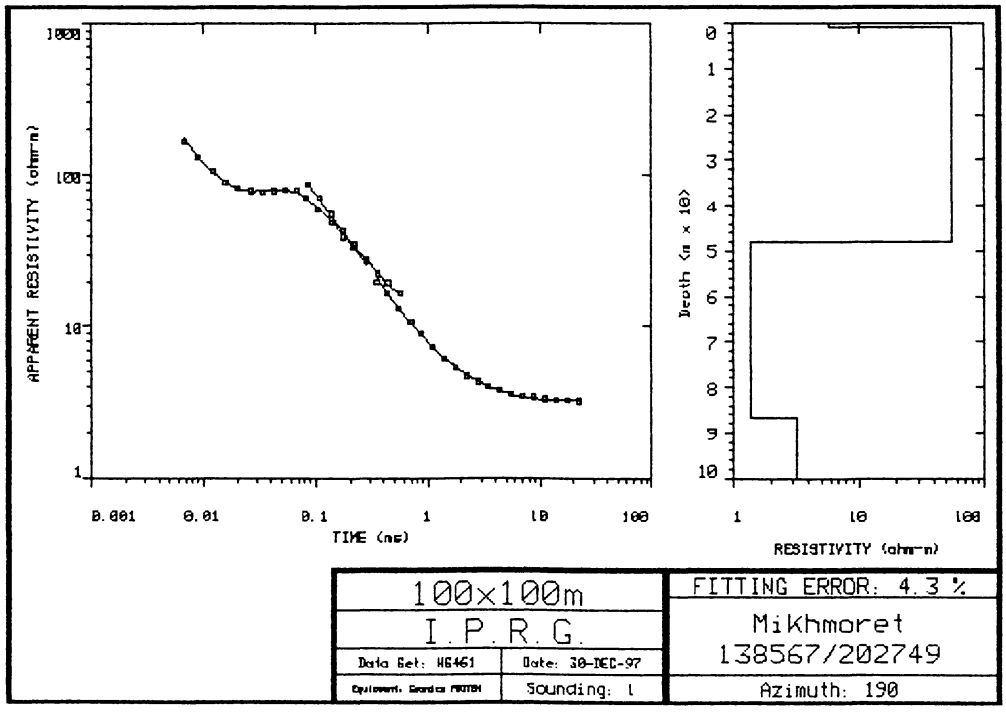
LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	7.92098E+00	1.67549E+00
2	8.92091E+01	6.27987E+01
3	2.48589E+00	2.12901E+01
4	1.13503E+01	8.30531E+01
5	1.37893E+00	



100×100m	FITTING ERROR: 5.5 %
I.P.R.G.	Mikhmoret 139275/202471
Data Set: HS460 / Date: 30-DEC-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 190

HS461

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	5.72423E+00	7.89502E-01
2	5.57902E+01	4.70539E+01
3	1.37302E+00	3.88999E+01
4	3.14806E+00	



100×100m	FITTING ERROR: 4.3 %
I.P.R.G.	Mikhmoret 138567/202749
Data Set: HS461 / Date: 30-DEC-97	
Equipment: Geonics PROTEM / Sounding: 1	Azimuth: 190

HS462

LAYER	RESISTIVITY (ohm-m)	THICKNESS (m)
1	1.73098E+00	8.42688E-01
2	1.04499E+02	4.29305E+01
3	1.48353E+00	4.50145E+01
4	2.90867E+00	

